Order Primates

A baculum is usually present except in Tarsiidae and Hominidae. Large primates produce only one young at a time but smaller species may produce two or more. Both lactation and gestation are long. Data on the reproductive physiology and endocrinology of primates are too extensive for thorough review in this context.

Reviews of cervix uteri (4189), anatomy (2625, 5842, 8616a), spermatogenesis (3993), reproductive cycles (6177, 10264a), copulation (2536, 2626), placentation (769, 6626), birth hour (5378), and litter size (6406) exist. Several journals are devoted solely to primates including: American Journal of Primatology, Contributions to Primatology, Folia Primatologica, International Journal of Primatology, Journal of Medical Primatology, Monographs in Primatology, and Primates.

Order Primates, Family Cheirogaleidae

Dwarf and mouse lemurs (*Allocebus* 100 g; *Cheirogaleus* 140-900 g; *Microcebus* 50-385 g; *Phaner* 350-500 g) are nocturnal, arboreal omnivores from Madagascar. The reproductive biology of *Allocebus* is unknown. The epitheliochorial and syndesmochorial, labyrinthine placenta has a central implantation into a bipartite or bicornuate uterus (4814, 8444, 9049, 10488, 10489) and is usually eaten after birth (3810, 8375, 8433, 8444). Estrus postpartum (3363) is reported. Neonates are furred and their eyes open at 1-4 d of age (3363, 3810, 3811, 4722, 6892, 8375, 8432, 8435, 8443, 8444). Sex of offspring may reflect maternal social or nutritive status (8379a). *Cheirogaleus* and *Microcebus* have 6 mammae (8443). Males have seminal vesicles, penial scales, and a baculum (5842, 8443, 8444). Details of female anatomy (4793, 8443, 8444), endocrinology (8376, 8377, 8378, 8379, 8382, 8454), and photoperiod (8446, 8447, 8449, 8450) are available.

NEONATAL MASS		
Cheirogaleus major	18.07 g; r17.2-19.5; n3 yg, n1 lt	8443, 8444
Cheirogaleus medius	18 g	4722
Microcebus coquereli	12 g	8435
Microcebus murinus	m: 2.7 g; n1	8443, 8444
	~4 g	4722
	f: 4.3 g; n1	8443, 8444
	f: 4.46 g; r3.5-7.0; n4	3810
	m: 6.46; r4-7.5; n7	3810
	6.8 g; sd1.3; n55	8375
Microcebus rufus	5-8 g	8435
NEONATAL SIZE		
Cheirogaleus major	HB: 76.7 mm; r73-82; n3	8443, 8444
Microcebus murinus	m: HB: 37 mm; n1	8443, 8444
	f: HB: 50 mm; n1	8443
LITTER SIZE		
Cheirogaleus major	mode 2; r2-3	8432, 8435
	3 yg; 1 lt	8443
Cheirogaleus medius	2.0 yg; r1-4; n32 lt, 9 f	3363
	2-3	4722
Microcebus coquereli	1.7; sd0.47; r1-2; n9	5192
	mode 2	8435
Microcebus murinus	1.83 yg; mode 2; r1-3; n12 lt	8443
	1.86 yg; mode 2; r1-3; n7 lt	1150, 8443, 8444
	2 yg	3810, 3811
	mode 2 emb, sometimes 3	8906
	mode 2; r2-3	8432, 8435
	2-3 yg	6892
	2-4; 1 rare	10282
	2.0; r1-3; n2 lt	4722
	2.17; mode 2; r1-3; n35 lt	8375
	3-4	8445
Microcebus rufus	2-3	8435
Phaner furcifer	1	1743, 8435

SEXUAL MATURITY		
Microcebus murinus	f: 1st mate: 1 yr	8435
	f: 1st birth: 11-12 mo	306
	f: 1st birth: 13 mo	3810
	f: 1st birth: 18 mo	8443, 8444
	m: testes large: 185.13 d; 95%ci 20.28	8450
	m: 1st sire: 11 mo	3810
	f/m: 1st mate: 7-8 mo	8432
	f/m: 1st mate: ~7-10 mo	8443, 8444
CYCLE LENGTH		
Cheirogaleus major	estrus: 2-3 d	8444
	~14 d	8445
	r23-31 d; n6 cycles, 3 f	8443
	25 d	8441
	~30 d	1150, 8443, 8444
Cheirogaleus medius	19.66 d; se1.58; r18-23; n9 cycles, 3 f	3363
Microcebus coquereli	16-46 d	5192
	~1 mo	8451
Microcebus murinus	estrus: 1 d	8435
	estrus: 2-3 d	8442, 8444
	estrus: 5.4 d; ±0.8; r3-13; n15	3810
	15-20 d	8442, 8445
	38.2 d; sd5.7; n4 isolated	8379
	40-55 d	8435
	mode 45-55 d 78%; r36-78; n23 cycles	1150, 8443, 8444
	50 d	6892
	mode 50-60 d; r35-108	306
	50.4 d; sd1.5; mode 39, 56; n34 cycles	3810
	53.7 d; sd5.9; n4 sensory cues (not tactile) from f	8379
	62.7 d; sd0.8; n4 tactile cues with other f	8379
Phaner furcifer	~15 d	8451
GESTATION		
Cheirogaleus major	70 d; n1	8432, 8443, 8444
Cheirogaleus medius	61.6 d; se0.96; mode 61 70%; r61-64; n10 lt, 6 f	3363
	70 d	4722
	70 d	8432
Microcebus coquereli	84-89 d	8435
	90.6 d; sd4.3; r85-100; n8	5192
	3 mo	8451
Microcebus murinus	3.5-4 wk	10282
	54-68 d	306
	58.3; sd0.4; n6	3810
	59-62 d; n7 lt	1150, 8443, 8444
	60.5 d; r59-62; n2	4722
	60.2 d; sd1.7; r56-64; n33	8374, 8375
LACTATION		
Cheirogaleus major	solid food: 25 d	8443, 8444
	nurse: ~1.5 mo	8443, 8444
Microcebus coquereli	wean: 134-138 d	5192
Microcebus murinus	emerge nest: 3 wk	6892
	solid food: 20-25 d	3810
	solid food: 3 wk	1150
	solid food: 1 mo	6892
	wean: 20-25 d	3810
	wean: 1 mo	8432
	wean: 40 d	8435
	wean: 1.5 mo	6892
	wean: 45 d	4722
	wean: 6 wk	1150, 8445
	independent: 60 d	4722
	wean: ~40 d	8443
Microcebus rufus	wean: 40 d	8435

INTERLITTER INTERVAL		
Microcebus murinus	71-75 d	8435
Microcebus rufus	71-75 d	8435
SEASONALITY		
Allocebus trichotis	birth: end Nov-Dec; native reports; Madagascar	7983
Cheirogaleus major	spermatogenesis: July-Oct; Madagascar	8444
	clitoris: open June, small Oct-June; n2; captive	8441
	mate: mid Nov; Madagascar	8435
	birth: Jan; Madagascar	8444
	birth: Jan; Madagascar	8432, 8435
	birth: Nov; captive, Madagascar	8445
Cheirogaleus medius	lethargy: Oct-Apr; captive	3363
	spermatogenesis: July-Oct; Madagascar	8444
	mate: Apr-June; captive	3363
	mate: Nov; Madagascar	4834
	preg f: Jan; n1; Madagascar	8445
	birth: Jan; Madagascar	4834, 8432
	birth: June-Aug; captive	3363
Microcebus coquereli	estrus: Apr, July; lab	8451
	mate: Apr; Madagascar	8435
	birth: July-Aug, Nov; lab	8451
Microcebus murinus	testes: increase Jan; n1; captive	8373
	testes: large Mar-June; captive	8445
	testes: large Oct, small July; s Madagascar	10282
	spermatogenesis: Oct, none July; few data; Madagascar	10282
	testosterone: peak Feb-Aug; captive	8377
	male repro: July-Jan; captive	306
	estrus: Mar-Aug; captive	8375, 8377
	estrus: Mar-Sept; captive, France	1150, 8442, 8446, 8448, 8451, 8453
	estrus: May-July; lab	6891
	estrus: Aug-Feb; captive	306
	estrus: Nov-Feb; captive, Madagascar	8442
	mate: Feb-Aug; captive, France	8379
	mate: Mar-July; captive	8378
	mate: Oct-Dec; Madagascar	8444
	mate: Sept-Jan; Madagascar	8432, 8435, 8445
	preg f: Oct-Nov; Madagascar	8906
	preg f: peak Oct-Nov; Madagascar	10282
	birth: Apr-May, July; captive, France	8444
	birth: May-June, Oct; captive	8375
	birth: May-Aug; captive, France	8443
	birth: May-Sept; zoo	4722
	birth: July; n1; lab	8451
	birth: Oct-Jan; Nosy Be I, Madagascar	8445
	birth: end Oct-early Feb; captive	306
	birth: Nov-Feb; Madagascar	8443, 8453
	birth: Nov-Mar; Madagascar	8432
	birth: end Nov-Feb; Madagascar	8444
	birth: Dec-Jan; ne Madagascar	8445
	repro: Oct-Feb; Madagascar	8446
Microcebus rufus	mate: Sept-Jan; Madagascar	8435
Phaner furcifer	testes: increase Jan-Mar; lab	8451
	estrus: mid Jan-June; lab	8451
	mate: mid Nov; Madagascar	8435
	birth: mid Nov; Madagascar	8440

Order Primates, Family Lemuridae

Lemurs (*Hapalemur* 0.8-2.5 kg; *Lemur* 1-3.7 kg; *Lepilemur* 500-900 g; *Varecia* 3-4 kg) are arboreal, often social, herbivores from Madagascar. The epitheliochorial, villous, diffuse placenta implants into a bipartite or bicornuate uterus (4770, 7563, 8443, 8444, 10310, 10962) and is sometimes eaten after birth (3362, 3538, 4718, 4722, 8433). The chorion produces a gonadotrophic complex (10636a). Neonates are furred with open eyes (3362, 3967, 4718, 4722, 8432, 8433, 8435, 8444). Females have 6 mammae (*Varecia* 8443). Nulliparous females may lactate (8369a). Males have a baculum and seminal vesicles (4793, 5842, 8435, 8443, 8444). Details of female anatomy (4793, 8443, 8444), endocrinology (305, 1045, 2636, 5525, 7957, 8371a, 8781, 9081, 9082, 9298, 9905, 9907, 11170, 11172), spermatogenesis (8959), birth (1440), and milk composition (1482) are available.

NEONATAL MASS		
Hapalemur griseus	32 g; n1	4722
Lemur catta	50 g; n1	4718
	r50-70 g; n10	768
	60-80 g	4722
	m: 85 g; n1	4718
Lemur coronatus	70 g	7947
Lemur mongoz	60 g	4718, 4722
Varecia variegata	80-85 g	7175
	81.5 g; r75-88; n2; died within 24 h	1127
	m: 84 g; n1	7947
	93.7 g; r73-111; n7	768
	93.9 g; r70-140; n59	1300
	100 g	8435
	mode 100 g; r60-125; <75 g premature; n34	768
	104.63 g; sd19.10; r70-125; n8	9906
NEONATAL SIZE		
Hapalemur griseus	HB: 60 mm	3967
	HB: 115 mm; n1	4722
Lemur coronatus	100 mm	7947
Lemur mongoz	138 mm; r136-140; n1	4718
Lepilemur mustelinus	HB: 100 mm	8443, 8444
LITTER SIZE		
Hapalemur griseus	1 yg	8435, 8438, 8439
	1; n1 lt	4722
	1 ea; n2	379
	2; n1	3967
Lemur catta	1	8432, 8433
	mode 1	6476
	1 yg; 2 occasional	8443
	1 yg ea; n10 lt	768
	1 ea; n37 lt	3340
	1.12; r1-2; n17 lt	4722
	1.15; r1-3; n80 lt	12128
	1.17 yg; mode 1; r1-3; n547 lt	4736
	1.18; r1-2; n22 f, 60 lt	8253, 11171
	1.2; r1-2; n9 lt	4718
	1.25; sd0.48; n56 lt	8936
Lemur coronatus	1 ea; n25 lt	3340
	1.5; r1-2; n2 lt	7947
	1.5; r1-2; n6 lt, 1 f	5523
	1.60; sd0.55; n5 lt	8936
Lemur fulvus	1	5573
	mode 1	6476
	1 yg; occ 2	8443
	1 emb ea; n3 lt	5575
	1 ea; n7 lt	3340
	1 yg ea; n8 lt	1230
	1 yg ea; n18 lt	1440
	1 ea; n19 lt	3340

	1.03 yg; n170 lt	4736
	1.06; sd0.28; n31 lt	8936
	1.06; sd0.25; n99 lt	8936
	1.12; sd0.33; n17 lt	8936
	1.45; sd0.68; n33 lt	8936
	1.46; sd0.67; n13 lt	8936
	2 emb; n1 f	1440
Lemur macaco	1	8432, 8433
	1 yg; 2 occasional	8443
	1 ea; n2 lt	1230
	1.02 yg; n82 lt	4736
	1.04; mode 1; r1-2; n24 lt	3340
	1.39; sd0.53; n44 lt	8936
Lemur mongoz	1	8432, 8433
	1 ea; n3 lt	8936
	1 ea; n4 lt	4718
	1 ea; n5 lt	4722
	1.08 yg; n110 lt	4736
	1.5; r1-2; n2 lt	4722
	2; n1 lt	4718
Lemur rubriventer	1	8432, 8433
Lemur rufipes?	1 emb ea; n4 lt	10962
Lepilemur mustelinus	1	8432, 8433
	mode 1	8444
	1 emb ea; n9 lt	5573
	1 yg ea; n12 lt	8443
Varecia variegata	1.3; mode 1; r1-3; n7 lt	3340
	1.5 yg; r1-2; n2 lt, n1 f	7947
	1.89; sd0.74; n38 lt	8936
	1.92 yg; mode 2; r1-3; n12 lt	1127
	1.95 yg; r1-3; n22 lt	4736
	2-4	9905
	mode 2; max 6	8435
	2.0 yg; mode 2; r1-4; n38 lt	3362
	2.16; mode 2; r1-6; n326 lt	1300
	2.22; sd0.81; n18 lt	8936
	2.47 yg; r1-3; n15 lt	768
	2.63 yg; sd0.52; mode 3; r2-3; n8 lt	9906
	2.64 yg; mode 3; r1-6; n89 lt	1300
	3; n1 lt	1044
	max 5	6476
	6 yg; n1 lt	5525
SEXUAL MATURITY		
Lemur catta	f: 1st conceive: 19.56 mo; sd5.57; r10-31; n9 f	11171
	f: 1st birth: 2.5 yr	4718
	m: 1st sire: 3 yr	4718
Lemur coronatus	f: 1st conceive: ~20 mo; n1	5523
Lemur macaco	f: 1st birth: 3 yr	8435
	f/m: 1.5-2.5 yr	8444
Lepilemur mustelinus	f/m: est 1.5 yr	8443, 8444
Varecia variegata	f: 1st estrus: 9 mo	1300
	f: 1st estrus: 20-22 mo	3362
	f: 1st conceive: 20.5 mo; r20-21; n2	1127
	m: testes descend: 16-22 mo	3362
	f/m: 20 mo	6476
CYCLE LENGTH		
Lemur catta	estrus: <24 h	4722
	estrus: 24 h	6225
	vaginal estrus: 3.67 d; ±0.37; r2-5	11172
	luteal phase: 25 d; n4 f	1045
	follicular phase: 14 d; n4 f	1045
	39 d; n4 f	1045
	39.3 d; se0.6; 76% r37-42; r33-45; n17 cycles	3111
	39.87 d; sd2.66; r32-45; n40 cycles, 16 f	11171

	40 d; between ovulations	6225
	6 wk	4722
Lemur coronatus	estrus: 1 d	5523
	34.2 d; sd5.8; r26-46; n12 cycles, 4 f	5523
Lemur fulvus	estrus: 24-48 h	1056
	estrus: 4 d; metestrus: 8 d; diestrus: 14 d; preestrus 3 d	2114
	estrus: 6.7 d; r4-10; n10 cycles, 2 f	2094
	26.9; se0.2; n38 cycles, 4 f no visual contact	1128
	30.2; se0.3; n30 cycles, 4 f in visual contact	1128
	28.3 d; se2.3; r24-33; n22 cycles, 6 f	2114
	30 d; n2 f	1056
Lemur macaco	estrus: 25 h	6225
	folllicular phase: 5 d; n6 cycles, 2 f	1045
	luteal phase: 26 d	6225
	luteal phase: 28 d; n6 cycles, 2 f	1045
	33 d; n6 cycles, 2 f	1045
	33.3 d; r32-35; n3 cycles	1044
	40 d	6225
Lemur mongoz	37.3 d; r29-48; n9	9527
Varecia variegata	estrus: 12-24 h	6476
	estrus: 24 h	6225
	mate: 1 d; n8	9906
	genital changes: 9.85 d; sd1.67; n8	9906
	follicular phase: 14 d; n3 cycles, 1 f	1045
	luteal phase: 24 d; n3 cycles, 1 f	1045
	luteal phase: 26 d	6225
	14.8 d; r8-22; n85 cycles	1300
	30 d; n8 cycles, 4 f	1127
	38 d; n3 cycles, 1 f	1045
	40 d; between ovulations	6225
	40-42 d; n3 f	3362
	40-44 d	9907
GESTATION		
Hapalemur griseus	140 d; n1	8438
Lemur catta	4 mo; n1	8443
	4 mo	5377
	133 d; r132-134; n2	4718, 4722
	134.5 d; r131-136; n8	3111
	135 d	6476
	135 d; n54 lt, 16 f	8936
	135.64 d; sd1.63; r134-138; n14 f	11171
	4.5 mo	8432, 8433, 8445
	144 d	4582
Lemur coronatus	122 d; n4 lt, 1 f	8936
	123.5; r123-124; n2	7947
	124.6 d; sd2.9; n6 lt	5523
Lemur fulvus	~15 wk; field data	5573
	117 d; n33 lt, 9 f	8936
	117 d; n89 lt, 19 f	8936
	120 d; n25 lt, 7 f	8936
	4 mo	1056
	4 mo	2153
	128 d	6476
	4.5 mo; n1	8443
Lemur macaco	125 d; n41 lt, 11 f	8936
	127 d; n2	4722
	127-129 d	3538
	128-129 d; n3	1044
	4.5 mo	8432, 8433
	4 mo 3 wk; r120-135 d; field data	8444, 8445
	143 d	4582

Lemur mongoz	114 d; n1	4722
	126 d	4718, 4722
	4.5 mo	8432, 8433
	137 d; r126-159; n4; 159 d prob includes 1 cycle	9527
Lemur rubriventer	4.5 mo	8432, 8433
Lepilemur mustelinus	~14 wk; field data; not many	5573
	120-150 d; field data	8444
	>4 mo	8445
	4-5 mo; est 5 mo	8443
	4.5 mo	8432, 8433
Varecia variegata	90, 90-100, 95-100, 105 d; n4 lt	1127
	100.75 d; sd1.16; r100-103; n8 lt	9905, 9906
	101 d; n1; induced ovulation	5525
	102 d; n1	1044
	102 d; sd1; r99-106; n14	1300, 7175
	102 d; n18 lt, 7 f	8936
	102 d; n35 lt, 11 f	8936
	102.5 d; ±1.5; mode 102-103; r99-106; n21 lt	3362
	103 d	7947
	104 d	6476
	~104 d; n1 twins; vaginal swelling to birth	5880
LACTATION		
Hapalemur griseus	wean: 6 mo	379
Lemur catta	solid food: 1 mo	4718
	wean: 5 mo	8433
	independent: 6 mo	8433
Lemur macaco	solid food: 1 mo	8433
	wean: 5 mo; r4-6	8433
	independent: 6 mo	8433
Lepilemur mustelinus	solid food: 45 d nibbles	8444
	solid food: 1.5-2.5 mo	8443
	nurse: 2.5 mo	8443
	wean: ≤4 mo	8443
	wean: min 4 mo	8444
LITTERS/YEAR		
Lemur catta	1	768
	1	6476
	mode 1	8432
Lemur macaco	1	8444
Lemur mongoz	1	4722
Lepilemur mustelinus	1	8432
	1	8444
SEASONALITY		
Hapalemur aureus	birth: Dec; Hanomefana, Madagascar	326
Hapalemur griseus	mate: Jan; captive	8438
	birth: Jan-Feb; ne Madagascar	8439
	birth: Sept; zoo	8445
	birth: Dec-Jan; Madagascar	8435, 8438
	birth: June; captive	8438
Lemur catta	testosterone: high Sept, Dec-Jan, low Apr; captive	3111
	mate: Mar-June; zoo	12128
	mate: end Mar-early July; s Madagascar	5573
	mate: Apr; s Madagascar	8445
	mate: end Apr; Madagascar	5377
	mate: end Oct-Nov; zoo	4722
	birth: Mar-June, peak Mar; n21/37; zoo Egypt	3340
	birth: Mar-July; zoo	4722
	birth: Apr-June; Madagascar	8433
	birth: Aug-Nov; Madagascar	8432

	birth: end Aug; Madagascar	5377
	birth: end Aug; s Madagascar	8445
	birth: end Aug-Nov; Madagascar	1430
	birth: end Oct; s Madagascar	5573
	birth: Sept; Madagascar	5376
	birth: Sept-Oct; n30/31; zoo	1230
Lemur coronatus	estrus cycles: end Nov-Mar; captive	5523
	birth: Apr-July, peak Apr-May; zoo, Egypt	3340
	birth: May; n2; zoo	7947
Lemur fulvus	estrus: Feb-mid May; ne Madagascar	5573
	mate: Apr; Madagascar	8444
	mate: Apr-June; ne Madagascar	8435
	mate: Aug, Oct-Apr, peak Dec; captive	2153
	mate: Nov-Apr; captive, France	2114
	mate: Nov-Dec; captive	6476
	preg f: July-Aug; ne Madagascar	5573
	preg f: end July-mid Aug; n4; Madagascar	5575
	birth: mid Feb-mid July, peak Apr-May; captive	2153
	birth: Mar-June; n29; zoo	12120
	birth: Mar-July; n26; zoo, Egypt	3340
	birth: end Mar-May; captive	6476
	birth: Aug, Nov; ne Madagascar	8435
Lemur macaco	mate: Apr-June; ne Madagascar	8435
	mate: mid Apr-June; Madagascar	8444
	birth: Mar-Apr; zoo	4722
	birth: Mar-June; n14; zoo, England	12120
	birth: Mar-July, peak Apr; n24; zoo, Egypt	3340
	birth: Apr-June; Madagascar	8433
	birth: Aug-Nov; Madagascar	8432, 8435
	birth: Sept; n2; zoo	1230
	birth: Sept-Oct; Nosy Be I, Madagascar	8445
	birth: Sept-Nov; Madagascar	8444
Lemur mongoz	mate: Dec; zoo	4722
	birth: Apr; zoo	4722
	birth: Apr-June; Madagascar	8433
	birth: peak Apr-June, also Jan, Mar, July; captive	9527
Lemur rubriventer	birth: Apr-June; Madagascar	8433
Lepilemur mustelinus	spermatogenesis: May-Aug; Madagascar	8444
	estrus: Feb-May; Madagascar	5573
	mate: Apr-July; Madagascar	8444
	mate: May-June; Nosy Be I, Madagascar	8445
	mate: May-July; Nosy Be I, Madagascar	8432
	preg f: June-Sept; Madagascar	5573
	birth: May-July; Madagascar	8433
	birth: Sept-Nov; Nosy Be I, Madagascar	8432, 8443, 8445
	birth: mid Sept-Oct; Madagascar	8444
	birth: Oct; n1; Madagascar	5573
Lepilemur ruficaudatus	estrus: May; lab, Madagascar	8451
	estrus: Nov, Jan; captive	8451
	birth: June; n1; lab	8451
Varecia variegata	testes: low summer; captive	3362
	estrus: June-Mar; zoo	9906
	mate: Dec-Jan; captive	6476
	birth: Mar-July, peak Apr >50%; zoo	1300
	birth: peak Apr-May; zoo	9906
	birth: Apr-June; n7; zoo, Egypt	3340
	birth: June; captive	6476
	birth: Sept; zoo	8445
	birth: Oct-Nov; e Madagascar	8432, 8435

Order Primates, Family Indriidae

Indris (*Indri* 7-10 kg), avahis (*Lichanotus* 600-1000 g), and sifakas (*Propithecus* 3-9 kg) are social, arboreal herbivores from Madagascar's rainforests. The epitheliochorial, diffuse placenta implants in a bicornuate or bipartite uterus (7563, 8444, 10962) and is eaten after birth (2854, 8433, 9096). Neonates are furred and open-eyed (8432, 8443, 8444, 9096). Males have a baculum and seminal vesicles (8443, 8444). Details of female anatomy (4793, 8443, 8444) and birth (9096) are available.

NEONATAL MASS		
Propithecus verreauxi	35-40 g; n1	115
NEONATAL SIZE		
Lichanotus laniger	HB: 90 mm	8443, 8444
LITTER SIZE		
Lichanotus laniger	1	8444, 8445
	1	8432, 8435
Propithecus verreauxi	1	4374
	1	8435
	1 ea; n3 lt	8443
	1-2	5376
SEXUAL MATURITY		
Propithecus verreauxi	f/m: 1st mate: 2.5 yr	5376
	f/m: est > 2.5 yr	8444
GESTATION		
Indri indri	17-22 wk; field data	8648
Lichanotus laniger	4-5 mo	7983
Propithecus diadema	est ~5 mo	8435
Propithecus verreauxi	~15 wk; field data; not many	5573
	est 4-5 mo	5575
	est 5 mo; n1; swollen genitals to birth	2854
	~5 mo	8435
	~5 mo; field data	8443, 8444, 8445
	~5 mo; field data	9095
	156.75 d; r154-160; n4	4374
LACTATION		
Indri indri	wean: 8 mo-< 1 yr	8648
Propithecus spp	solid food: 53-57 d	8432
Propithecus verreauxi	solid food: 1-2 wk	9096
	solid food: 7 wk	4374
	solid food: 11.5 wk	2854
	wean: < 5 mo	9096
LITTERS/YEAR		
Lichanotus laniger	1	8444
	1	8432
Propithecus verreauxi	1	8445
SEASONALITY		
Indri indri	mate: Jan-Mar; e Madagascar	8648
	birth: end May; e Madagascar	8648
Lichanotus laniger	birth: Aug-Sept; ne, e Madagascar	8432, 8435
	birth: end Aug; ne Madagascar	8444, 8445
	birth: Oct-Nov; e Madagascar	8444, 8445
Propithecus diadema	birth: Sept-Oct; e Madagascar	8435
Propithecus verreauxi	estrus: mid Nov-Mar; nw Madagascar	5573
	mate: Jan-Mar; nw Madagascar Madagascar	8444, 8445
	mate: Jan-Mar; w,s,e coast Madagascar	8435
	mate: Feb; Madagascar	5376
	mate: Mar; nw, s Madagascar	9095
	preg f: end rains; none June-Dec; n Madagascar	5575
	birth: May-Aug; w,s,e coast Madagascar	8435
	birth: June-July; nw Madagascar	5573
	birth: end June-mid July; Madagascar	8444, 8445
	birth: early-mid July; Madagascar	5376, 5377
	birth: Aug; nw, s Madagascar	9095

Order Primates, Family Daubentoniidae, Genus *Daubentonia*

The rare, nocturnal aye-aye *Daubentonia madagascariensis* from Madagascar eats fruit, flowers, and insect larvae. Males of this 2.5-3 kg, solitary and arboreal omnivore have scrotal testes, extremely small seminal vesicles, large Cowper's glands, and a baculum (5572, 8163, 8444). Females have 2 inguinal teats (8435, 11821), a short clitoris, long perineum, and scrotal-like appendages lateral to the clitoris (4789). The villous, laminar, epitheliochorial, diffuse placenta (4765, 7563) implants into a bicornuate uterus (4765). Details of anatomy (8177, 8443, 8444) are available.

LITTER SIZE		
Daubentonia madagascariensis	1	8432, 8434, 8437
LACTATION		
Daubentonia madagascariensis	solid food: 3 mo; n1	11821
	wean: 6-7 mo; n1	11821
INTERBIRTH INTERVAL		
Daubentonia madagascariensis	2-3 yr	8437
SEASONALITY		
Daubentonia madagascariensis	birth: Feb-Mar; Madagascar	8432
	birth: Oct-Nov; Madagascar	8434, 8437

Order Primates, Family Lorisidae

Lorises (*Arctocebus* 150-450 g; *Loris* 100-350 g; *Nycticebus* 375-2000 g; *Perodicticus* 850-1600 g) are nocturnal, arboreal omnivores from India, Africa, and southeast Asia. The epitheliochorial, villous, diffuse placenta has a superficial circumferential implantation into a bicornuate uterus (4758, 4759, 4771, 4774, 7563, 7567, 10310) and is eaten after birth (5478). A vaginal plug is reported (8888). A postpartum estrus may occur if the neonate dies (1739, 5195, 5198, 5309, 6767, 8888). Neonates are furred and open-eyed (34, 974, 1736, 2092, 4004, 5758). Females have 2-6 mammae (1736, 4774, 5198, 7161) and pseudoscrota (5758). Males have abdominal testes, seminal vesicles, and a baculum (5842, 8443, 8880, 8881, 8886). Details of female anatomy/oogenesis (249, 1737, 2290, 2774, 2776, 3733, 5160, 8925), endocrinology (8371a), and birth (1736) are available.

NEONATAL MASS		
Arctocebus calabarensis	27 g; r24-30; n2	1736, 1742
	f: 37 g; n1	5309
Nycticebus coucang	33.0 g; n1 stillborn	5198
	50 g; n1	34
	f: 50.75 g; r50.5-51.0; n2	5198
	m: 51.57 g; r43.5-57.0; n3	5198
	60 g; n1, 3 d	4785
Perodicticus potto	28 g; n1	2109
	35.7 g; r30-42; n6	2092, 2095
	f: 52 g; n1	1738, 1742
NEONATAL SIZE		
Perodicticus potto	11.75 mm; n1	2109
LITTER SIZE		
Arctocebus calabarensis	1 yg ea; n3 lt, 2 f	1736
Loris tardigradus	1 yg ea; n9 lt	5195, 8937
	1 yg; 2 very occasional	8501
	1+; 2 occasional	8880
	1.5 emb; r1-2; n2 lt	4771
	1.6 emb; r1-2; n87 f	8888
Nycticebus coucang	1 yg	2313
	1	8619
	1 yg; n1 lt	34
	1 yg; n3 lt	4785
	1 yg ea; n19 lt	5198, 8937
	1.25; mode 1; r1-2; n8 lt	2109
	1.3; r1-2; n6 lt	851

Perodicticus potto	1; n1 lt	974
	1; n1 lt	11401
	1; r1-2; 2 rare	5758
	1 ea; n7 lt	2109
	1.2 yg; r1-2; n5 lt	2095
SEXUAL MATURITY		
Arctocebus calabarensis	f/m: ~9-10 mo	1742
Loris tardigradus	f: 1st estrus: 10 mo; n1	5195
	f: 1st estrus: 340 d; sd56.5; n4	8937
	f: 1st conceive: 364 d; n1	5195
	f: 1st repro: 2 yr	8881
	m: 1st spermatogenesis: 10-11 mo	8881
Nycticebus coucang	f: 1st estrus: 19 mo; r17-21; n2	5198
	f: 1st mate: 20.3 mo; r18-23; n3	5198
	f: 1st birth: 3 yr; n1	5198
	m: 1st sire: 17 mo; n1	5198
Perodicticus potto	f/m: 9 mo	5758
	f/m: ~18 mo	1742
CYCLE LENGTH		
Arctocebus calabarensis	39 d; r36-45; n4 cycles, 1 f	6767
Loris tardigradus	33.6 d; sd4.5; r29-40; n3 f	5195
Nycticebus coucang	36.4 d; sd4.5; r29-45; n28 cycles, 10 f; interestrus interval	5198
	42.3 d; mode 40; r37-54; n17 cycles, 2 f	6767
Perodicticus potto	37.9; se~1.0; n36 cycles, 2 f; vaginal smears	5159
	estrus: 1.8 d; metestrus: 4.6 d	
	diestrus: 27.8 d; proestrus: 3.7 d	5159
	38.3 d; r21-48; n37 cycles, 3 f	6767
	50 d	8443
GESTATION		
Arctocebus calabarensis	134 d; r131-136; n3	6767, 6768
Loris tardigradus	163 d; r160-166; n2	6767, 6768
	167.2 d; sd1.5; r166-169; n4	5195, 8937
	174 d; n1	7881
Nycticebus coucang	174 d	11564
	192.2 d; sd3.6; r185-197; n9 lt, 5 f	5198, 8937
	193 d; n1	6767
Perodicticus potto	193 d; n1	1738, 1742
LACTATION		
Arctocebus calabarensis	solid food: 1.5 mo	1736
	wean: ~100-130 d	1742
Loris tardigradus	wean: 5 mo; from graph	8881
	milk present: 169 d; sd32.0; n5	5195, 8937
Nycticebus coucang	milk present: 180 d; sd11.3; n2	8937
	wean: ~6 mo; n5 lt, 4 f	5198
	wean: min 9 mo	2109
Perodicticus potto	solid food: 5 wk	5758
	wean: 120-180 d	1742
	wean: 2-3 mo	1740
INTERBIRTH INTERVAL		
Arctocebus calabarensis	131 d; n1	5309
	4.5-5.0 mo; n1	1739
Loris tardigradus	9.5 mo	5195
Nycticebus coucang	16.2 mo; r9-38; n7 f	5198
Perodicticus potto	min 6.5 mo	2092
SEASONALITY		
Arctocebus calabarensis	birth: Nov-May; Rio Muni, equatorial Guinea	8507
	birth: yr rd, low June-Aug; n33; Gabon	1742
Loris tardigradus	epididymal sperm: yr rd; Bangalore, India	8880, 8881
	accessory glands: peak May; Bangalore, India	8881
	estrus: June-July, Sept-Nov; Bangalore, s India	8886, 8888
	mate: June-July; Bangalore, India	8881
	term emb: Mar-May, Nov; Bangalore, s India	8888
	birth: Apr-May, Nov-Dec; Bangalore, s India	8880
Nycticebus coucang	preg f: yr rd, peak May; n146; captive	12120

	birth: Nov-Apr; n16/20; captive	851
	birth: Mar-May; n12/19; captive	5198
Perodicticus potto	testes: large Mar-May; no data after May; Africa	1739
	birth: Apr-May; n5; Nov-Feb; n9; e Africa	5758
	birth: June; n1; Zaire	8854
	birth: Aug-Jan; n17; Gabon	1739, 1742
	birth: not seasonal; n7; zoo	2109
	lact f: Jan; n1; s Cameroun, Benito, Equatorial Guinea	647
	lact f: Feb; n1; Kenya	4926

Order Primates, Family Galagidae

Bush babies (*Galago* 45-360 g; *Otolemur* 0.6-2 kg) are nocturnal, arboreal omnivores from African forests. Males are larger than females (4361, 4362, 5758, 7770). Oogenesis occurs in adults (1496, 1498, 3728, 3732, 3733, 4665, 5160). The endotheliochorial and epitheliochorial, diffuse placenta has a central, superficial, often antimesometrial implantation into a bicornuate uterus (1489, 1491, 1493, 1494, 1495, 1497, 2479, 3731, 3732, 4761, 4764, 5732, 7563, 8221, 8222, 10154, 10310) and is eaten after birth (2714). A postpartum estrus occurs in *Galago*, but not as regularly in *Otolemur* (1490, 2712, 2713, 2714, 4631, 5758, 5936, 9507, 11062, 11063, 11171). Neonates are furred and open eyed (973, 1491, 2711, 2788, 4122, 5758, 5936, 6444a, 6615, 8443, 9507, 10158). Females have 6 mammae (*Galago* 331, 1491, 1493; cf 10158). Males have seminal vesicles (1491) and androgen-dependent penial spines (2619, 2637). Details of anatomy (1491, 4211), gonadal development (12031), endocrinology (1444, 2268, 2636, 2872, 5194, 11170), birth (2714), milk composition (8541), and reproductive energetics (9511) are available.

NEONATAL MASS		
Galago alleni	24 g	1742
Galago demidovii	5-10 g	1742
	11.6 g; n1	2788
	25 g	5758
Galago senegalensis	9.3-9.6 g; near term	10154
	m: 11.45 g; n23	2712
	11.50 g; r8.5-15.5; n42; 12.11 g; n10 singletons; 11.31 g ea; n32 twins	2712, 2714
	f: 11.56 g; n19	2712
	f: 11.7 g; r8.5-14.8; n28	2711
	m: 11.9 g; r8.6-15.5; n28	2711
	15 g; n2	5936
	15.6-22.6 g; single births	2712, 2714
Galago zanzibaricus	14.1 g; r13-15; n3	4122
Otolemur crassicaudatus	40 g	5758
	47.4 g; sd6; twins each 45.9 g; n16	11062, 11063
	63 g; r46.4-72.7; n3	7870
	min 72.5 g; n5	11575
NEONATAL SIZE		
Galago demidovii	75 mm	5758
Galago senegalensis	58 mm; n2	5936
Otolemur crassicaudatus	101.0 mm; r81-108; n3	7870
	min 112 mm; n5	11575
WEANING MASS		
Galago senegalensis	90-110 g; mean +- 1sd; from graph; 70 d	2711
LITTER SIZE		
Galago demidovii	1.2 emb; r1-2; n18 f	1742
	1.3; r1-2; n3 lt	2788
	1.5; r1-2; n10 lt	5758
	2; n1 lt	973
	2; n1 lt	8443
	2.0 emb; n3 lt	8221
Galago elegantulus	1 emb ea; n9 f	1742
Galago inustus	1 emb; n1 f	5758
Galago senegalensis	1-2	9507
	1; north; 2; south	5758

	1 emb ea; n5 f	4926
	1 ea; n6 lt, 1 f	9137
	1.03; r1-2; n36 f	1488, 1491, 1493, 1495
	1.25 emb; r1-2; n4 lt	4665
	1.55 yg; sd0.50; n42	5196, 8937
	1.62; r1-3; n29 lt	2712
	1.875 yg; r1-3; n8 lt, 4 f	2714
	2	8443
	mode 2; n>11 lt	7769
	2; n1 lt	6615
Galago zanzibaricus	1 yg	7769
	1-2	5758
	1.29; r1-2, n7 lt	4122
Otolemur crassicaudatus	1-3	5196
	1; n1 lt	2109
	1 yg; n1	2334
	1 ea; n2 lt, 1 f	8443
	1.03 yg; r1-2; n86 lt	2855
	1.11 yg; mode 1; r1-2; n201 lt	4631
	1.12; r1-2; n69 lt	11062, 11063
	1.14; r1-2; n180 lt, 42 f	2637, 8253, 11171
	1.6 yg; se0.09; r1-3; n60 lt	2855
	1.64 yg; sd0.63; n77	5196, 8937
	2	11253
	2-3 88-100%; r1-3	7769
	2.0; mode 2; r1-3; n8 lt	681, 2713
SEXUAL MATURITY		
Galago alleni	f/m: 8-10 mo	1742
Galago demidovii	f/m: 8-10 mo	1742
Galago senegalensis	f: 1st estrus: 8 mo	5758
	f: 1st estrus: 265 d; sd95.6; n17	8937
	f: 1st conceive: mode 200 d; min 120	2711
	f: 1st mate: ~20 mo	1495
	f/m: 11 mo, 324 d; n1	2268
Galago zanzibaricus	f/m: 1 yr	5758
Otolemur crassicaudatus	f: 1st estrus: 445 d; sd89.8; n8	8937
	f: 1st mate: ~1 yr	1439
	f: 1st mate: min 16 mo; mode 19 mo	2855
	f: 1st mate: min 22 mo; mode 26.5 mo	2855
	f: 1st conceive: 1.89 yr; sd0.85; r0.99-4.58; n20	11171
	f: 1st conceive: 2.52 yr; sd0.97; r1.45-3.83; n5	11171
	m: 1st sire: min 10 mo; mode 18.5 mo	2855
	m: 1st sire: min 22 mo; mode 26 mo	2855
	f/m: 1.75 yr	1848
CYCLE LENGTH		
Galago senegalensis	estrus: 48 h	1495
	estrus: 3-6 d	6615
	estrus: 3-6 d	8442
	estrus: 4.8-6.7 d; n11 f	2268
	estrus: 7 d	9507
	14 d	8442
	28-30 d	2714
	4-6 wk	1490, 1491
	31.7 d; mode 31; r29-39; n31 cycles, 4 f	6767
	32.9 d; se0.6; r29.2-39.3 per f; n168 cycles, 11 f	2268
	39 d; r36-42; n3 cycles, 1 f	8443
	6 wk	9507
Otolemur crassicaudatus	estrus: 5.8 d	2870, 2872
	luteal phase: 24 d	2872
	39 d; se4.5; mode 34-38; n79 cycles	4631
	43.6; sd4.56; r39-49; n5 f	2637
	44 d; sd8.0; r35-59; n10 cycles	2870, 2872
	50.3 d; mode 17; r3-104; n54 cycles	11062, 11063

GESTATION		
Galago alleni	133 d	1742
Galago demidovii	1-3.7 mo; n1; pairing to birth	1589
	6 wk	5541
	3-4 mo	5758
	111-114 d	1742
Galago senegalensis	4 mo	8443
	4 mo; field data	9507
	~4 mo	6615
	~4 mo; field data	1488, 1490
	122-125 d; n4 lt, 2 f	2714
	123-124 d	5936
	123.5 d; r121-124 (± 1 d); n13	2712
	124.5 d; sd2.4; n28	5196, 8937
	139 d; ±3; n30	2268, 2714
	145 d; r144-146; n4	6767
Galago zanzibaricus	120 d; n4	4122
Otolemur crassicaudatus	125.2 d; se2.69; r121-133; n4	2855
	128 d; sd13; n69	11062, 11063
	18.5 wk	1439
	4 mo 10 d; n2	8443
	130.2 d; sd2.5; n12	5196
	131.9 d; sd3.4; r128-134; n20	4631
	132.2 d; sd4.76; r126-138	2637
	132.8 d; sd2.6; r130-136; n5	2872
	133 d	2713
	133 d; ±2	7870
	135.6 d; sd3.7; n21	5194, 5196, 8937
LACTATION		
Galago demidovii	solid food: 2 wk	1589
	suckle: min 5 wk	1589
	wean: 40-50 d	1742
	wean: ~6 wk; n5	5758
Galago senegalensis	solid food: 4 wk	10158
	solid food: 4 wk	9507
	solid food: 2 mo	5758
	nurse: 14 d; 1st decline; 45 d; 2nd decline	2711
	nurse: ~6 wk	5758
	nurse: 10-11 wk	2711, 2713
	nurse: 100-111 d	5192
	nurse: 100.3 d; sd9.9; n44	5193
	nurse: 101 d; sd11.1; n28	8937
	nurse: 3.5 mo	6615
Galago zanzibaricus	solid food: 1 mo	5758
	solid food: > 1 mo	4122
Otolemur crassicaudatus	solid food: mode 8 d	9507
	solid food: 1 mo	5758
	wean: begins 21 d	10158
	nurse: ~10 wk	10158
	nurse: ~3+ mo	5758
	nurse: 134 d; sd10.5; n9	8937
	nurse: 136.4 d; sd11.4; n19	5192, 5193
	nurse: 139.9 d; sd16.3; n17	5192, 5193
INTERLITTER INTERVAL		
Galago senegalensis	median 189 d; r124-325; n22 young die	5197
	7.5 mo	2712
	median 259.5 d; r167-454; n38 young live	5197
Otolemur crassicaudatus	min 119 d	2855
LITTERS/YEAR		
Galago demidovii	1	1742
	2?	11253
Galago zanzibaricus	mode 2	7769
	2	4122

SEASONALITY

Galago alleni	mate: yr rd; Cameroun	9468
	birth: Jan-Mar, May, July-Aug, Nov-Dec; n10; Gabon	1742
Galago demidovii	testes: mod-small Dec; Uganda	5758
	mate: end May; Katanga, Zaire	3732
	mate: yr rd, peak Aug; Uganda	5758
	birth: peak Jan-Apr, low June-July; Gabon	1742
	birth: Oct-Nov; Zaire	11253
	birth: yr rd, peaks Dec-Feb, June; Uganda	5758
Galago elegantulus	mate: low June-Oct; Cameroun	9468
	birth: Sept-Feb, Apr-May; n11; Gabon	1742
Galago inustus	testes: small July; n1; central Africa, s,w Uganda	5758
	mate: peak Nov-Dec; central Africa, s,w Uganda	5758
	preg f: Feb; n1; central Africa, s,w Uganda	5758
Galago senegalensis	anestrus: mid Feb; Tanzania	186
	estrus: Nov-Jan, Aug; 3.5 mo anestrus; Sudan	1491
	estrus: yr rd; Sudan	1490
	estrus: yr rd; captive	5197
	mate: Mar-Sept; captive	2714
	mate: end May; Katanga, Zaire	3732
	mate: end May-July, end Sept-Oct; S Africa	2713
	preg f: Dec-Jan, Mar-Apr, Oct; n24; Sudan	1488, 1491, 1495
	preg f: Feb, Nov-Dec; n5; Kenya	4926
	preg f: Sept; Rwanda	4665
	birth: Jan-Feb, mid Sept-Nov; Transvaal, S Africa	7769
	birth: Mar-Apr, June, Dec; n4; Sudan	1491
	birth: Apr; n1; captive	6615
	birth: July-Jan; captive	2714
	birth: Aug-Dec, peak Sept-Oct; Zambia, s Tanzania	5758
	birth: mid Sept-mid Nov; n5; zoo	1230
	birth: Oct, Feb-Mar; n Transvaal, S Africa	2713
	birth: Oct-Nov, Jan-Mar, peak Feb; S Africa	681
	birth: yr rd; Botswana	10154
	birth: yr rd; captive	5197
Galago zanzibaricus	preg f: Feb, June-Dec; Kenya	7769
	birth: Feb-Mar, Aug-Oct; Kenya	7769
Otolemur crassicaudatus	mate: end May; Katanga, Zaire	3732
	mate: June-July; Transvaal, S Africa	8965
	mate: est June-July; ne Transvaal, S Africa	2713
	birth: Mar; Somalia	5758
	birth: May; n1; zoo	2109
	birth: Aug-Sept; Zambia, Zimbabwe	10158
	birth: est Aug-Oct; few data; Kenya	7769
	birth: Aug-Nov; Tanzania, Zambia	5758
	birth: Oct-Nov; Zaire	11253
	birth: Oct-Nov; S Africa	1848
	birth: Oct-Nov; Transvaal, S Africa	7769
	birth: Nov, S Africa; ne Transvaal, S Africa	2713, 681
	birth: Nov; Transvaal, S Africa	8965
	birth: yr rd; captive	11063

Order Primates, Family Tarsiidae, Genus *Tarsius*

Tarsiers (*Tarsius bancanus*, *T. spectrum*, *T. syrichta*) are small (80-160 g), arboreal insectivores from tropical forests of southeast Asia. They are solitary except for reproduction (3364, 6310). Copulation lasts 60-90 sec with multiple thrusting and does not result in a copulatory plug (11943; cf 4788a). The 1.2-2.5 g, deciduate, hemochorial, labyrinthine, discoidal or diffuse placenta has a superficial, mesometrial implantation into the bicornuate or bipartite uterus (231, 1491, 4113, 4757, 4759, 5199, 5733, 6124, 7563, 11036, 11164). Neonates are mobile, furred, open-eyed, and up to 20% of maternal mass (557, 4374a, 6310, 11613). Females have 6 mammae (4788a). Male seminal vesicles are large (4788a). Details of anatomy (4788a, 11910a) are available.

NEONATAL MASS		
Tarsius bancanus	28.5 g; n1	5199
Tarsius spectrum	24 g	6310
	~28 g	557
Tarsius syrichta	23.2 g; r20-31.5; n9	4374a
	f: 25.3 g; n1	11036
	m: 27.2 g; n1	11036
NEONATAL SIZE		
Tarsius syrichta	m: 62 cm; n1	11036
	f: 66 cm; n1	11036
LITTER SIZE		
Tarsius bancanus	1	5192, 5199
	1	3364
	1 emb ea; n2 f	2313
	1 emb ea; n2 lt	2313
Tarsius spectrum	1	557
	1	6310
	1 emb ea; n5 lt	10482
Tarsius syrichta	1	11613
	1 yg; 1 lt	9647
CYCLE LENGTH		
Tarsius spp	23.5 d; ±0.7; r23-28; n1 f	1662
Tarsius bancanus	estrus: 2.2 d; r1-3	11942, 11943
	genitalia swollen: 6-9 d	11942, 11943
	22.5 d; ±3	5192
	24.0 d; sd3.2; r18-27; n6; smears	11942
	25.9 d; sd3.2; n9; visual swellings	11942
GESTATION		
Tarsius bancanus	178 d	5199
Tarsius spectrum	~6 mo	557
Tarsius syrichta	178 d	4374a
	est 6 mo	11036
LACTATION		
Tarsius spectrum	solid food: 3 wk	6310
INTERBIRTH INTERVAL		
Tarsius spectrum	262.4 d; r187-406; n5 intervals, 3 f	4374a
	1 yr	6310
SEASONALITY		
Tarsius bancanus	birth: Mar-July; Borneo, Indonesia	3364
	preg f: Aug; n2; N Borneo, Indonesia	2313
Tarsius spectrum	mate: yr rd; n88; captive	12120
	conceive: Oct-Dec; Borneo, Indonesia	6310
	preg f: yr rd, peak Oct-Nov; n1000+; Indonesia	11164
	preg/lact f: Oct-Feb; Borneo, Indonesia	6310
Tarsius syrichta	birth: Jan-Feb, Apr-May, Aug, Oct-Nov; n9; captive	4374a

Order Primates, Family Callithricidae

Marmosets and tamarins (*Callithrix* 190-600 g; *Cebuella* 100-150 g; *Leontopithecus* 360-800 g; *Saguinus* 225-900 g) are monogamous or social, arboreal omnivores from the forests of tropical Central and South America. Only dominant members of a group breed (*Callithrix* 3, 5, 8, 9314; *Saguinus* 2368, 3050, 3055, 3605, 4590, 9512, 11664; cf 10749) and a concealed estrus is reported (*Leontopithecus* 10498; *Saguinus* 1235). The 5-10 g, hemochorial, labyrinthine, bidiscoidal placenta (215, 770a, 1687, 4113, 4310, 4769, 4842, 5375, 7497, 11848, 11852, 11855, 11966) secretes gonadotrophins (10636a) and is shared by twins (767, 5854). Placentophagia occurs (1753, 6620, 6647, 6813, 8253, 9313, 9315, 10388). A postpartum estrus within 1-2 wk is usual (1686, 2366, 3049, 3120, 4540, 4544, 4545, 4549, 4550, 5850, 8487, 9313, 9314, 10184, 10185, 12098; cf 2153, 3503, 10199, 12098). Neonates are furred with open eyes (1753, 1821, 3049, 6222, 6621). Females have 2 mammae (*Callithrix* 5147). Males and siblings contribute to parental care by carrying young (*Callithrix* 4540, 4545, 6621, 6813, 10673, 11577; *Leontopithecus* 4838, 10184; *Saguinus* 3051, 5584, 6477, 7583, 10673, 11577, 11594, 11733).

Details of anatomy (10337, 11847, 11851), endocrinology (4306; *Callithrix* 2, 3, 5, 8, 1167, 1686, 2623, 2624, 2630, 2635, 2869, 3119a, 4307, 4308, 4372, 4378, 4379, 4380, 4540, 4542, 4545, 4549, 4550, 4566, 4567, 4568, 4844, 4858, 4859, 4862, 4864, 4865, 4865a, 4867, 4868, 5664, 5719, 5720, 5927, 6646, 7110, 8750, 9050, 9813, 10551, 11517; *Cebuella* 12100a; *Leontopithecus* 3507, 3508, 3509, 5859; *Saguinus* 1234, 1235, 3054, 3055, 3055a, 3504, 3505, 3694, 4590, 4591, 4864, 4866, 8748, 8752, 8767, 10671, 12098, 12099, 12100), sexual development (2, 6, 3052, 3053, 3397, 3695, 10671, 12099), spermatogenesis (4203), birth (10389), and reproductive energetics (5819) are available.

NEONATAL MASS

Callithrix jacchus	14.5 g; n34 twins, 17 lt	6647
	18.6 g; n36 triplets, 12 lt	6647
	22 g; n2	10388
	23.6 g; r18-27; n4 ea triplets	4306
	25 g; from graph	2
	25-30 g	4082
	25-32 g; varies inversely with litter size	1687
	28 g; n > 25	4544, 4545
	28.9 g; r27.5-30.0; n3 triplets, 1 d preterm	3049
	30 g; n1	6620, 6621
	30.3 g; r21-38; n9 ea twins	4306
	32.5 g; sd3.7; n3 singletons	6647
	34.7 g; n1	3049
Cebuella pygmaea	14-16 g	1821, 1822
	27 g; n1 stillborn	6222
Leontopithecus rosalia	49.6 g; r44.0-53.3	11749
	51.9 g; r50.2-53.65; n2, 1 d	2792
	m: 57 g; n1	215
	60.6 g; sd6.22; r52.1-74.6 g; n11; from graph	4838
Saguinus nigricollis	43.5 g; n10	11875
Saguinus oedipus	f: 30.5 g; r25.0-42.0; n2 triplets	11874
	34.3 g; r32-40; n3 ea triplets	4306
	36.1 g; r25-45; n20 ea twins	4306
	37.0 g; 34-40; n4 ea singletons	4306
	m: 38.2 g; r31.2-43.5; n6 triplets	11874
	41 g; n1	3049
	f: 42.1 g; r35.7-52.0; n12 twins	11874
	m: 44.0 g; n1 singleton	11874
	m: 44.0 g; r35.3-50.8; n20 twins	11874
	f: 46.4 g; r36.5-55.8; n1 singleton	11874

NEONATAL SIZE

Callithrix jacchus	CR: 68.6-91.8 mm; n14 neonates	8486
	CR: 70-72 mm; 146-149 d of gest	8486
	CR: 78.3 mm; r72-84; n6 ea triplets	4306
	80 mm	4545
	CR: 84.1 mm; r76-89; n11 ea twins	4306
Leontopithecus rosalia	HB: 100-114 mm	5854
	m: 122 mm; n1	215
Saguinus oedipus	CR: 89.3 mm; r87-91; n4 ea triplets	4306
	CR: 92.0 mm; r82-103; n20 ea twins	4306
	CR: 95.5 mm; r94-97; n4 ea singletons	4306

WEANING MASS

Callithrix jacchus	~60 g; 4 wk	5768
	~75 g; 6 wk	5768
	~125 g; 12 wk	5768
Saguinus oedipus	~70 g; 4 wk	5768
	~100-110 g; 7-8 wk	5768

LITTER SIZE

Callithrix spp	1.86; r1-3; n250 lt	4309
Callithrix argentata	1.5 yg; r1-2; n2 lt	4306
	1.83; mode 2; r1-2; n12 lt	4313
	2	10390
	2; n3 lt	5767
	2 yg ea; n7 lt	5768

Callithrix jacchus	1.33; r1-2; n3 lt, 1 f	1753
	1.54; r1-3; n41 lt	3697
	2	3
	mode 2; r1-3	1687, 4541, 4544, 4549, 7497
	2; n1 lt	6813
	2.0; r1-3; n2 lt	9313
	2.0 yg; r1-3; n14 lt	6620, 6621
	2.0 yg; mode 2; r1-3; n144 lt	4306, 4313
	2+; mode 2; r1-3	4545
	2.03 yg; mode 2; r1-4; n115 lt	5768
	2.05 yg; 1.88 alive; n66 lt, 30 f	10675
	2.05; r1-3; n87 lt	5767
	2.1; mode 2; r1-2	4551
	2.18; mode 2; r1-3; n17 lt	4550
	2.2 yg; mode 2; r1-3; n124 lt	1468
	2.22 yg; mode 2; r1-3; n159 lt	8487
	2.3; r2-3; n3 lt	12128
	2.33 yg; 1.7 alive; n6 lt, 2 pr	11874
	2.33 yg; 1.71 alive; mode 2; r1-3; n24 lt	5147
	2.42 yg; mode 2-3; r1-4; n50 lt	9315
	2.48 yg; mode 3; r1-4; n202 lt	9316
	2.5 yg; r2-3; n14 lt; triplets not raised	10388
	2.51 born; mode 3; r1-4; n87 lt	1468
	2.57; r2-3; n14 lt	3120
	2.61 yg; r1-4; n114 lt	9317
	2.68; mode 3; r1-4; n129 lt	8663
	2.69 yg; mode 3; r1-4; n202 lt; usually 2 nursed	1167
	2.82 yg; r2-4; n28 lt; no more than 2 yg reared	10391
Cebuella pygmaea	1.78; mode 2; r1-3; n9 lt	4306, 4313
	1.85 yg; mode 2; r1-3; n41 lt	12100a
	mode 2	10198
	2; n1 lt	5767
	2 yg; n1 lt	8891
	2 yg; n1 lt	5768
	2.02 yg, 1.74 alive; mode 2; r1-3; n39 lt	6222
	2.1 yg; r2-3; n8 lt, 3 f	1821, 1822
Leontopithecus rosalia	1.65 yg; n23 lt	2579a
	1.8; 2.1; n2 samples	5856
	1.9; mode 2; r1-3; n11 lt	4838
	1.94; mode 2; r1-3; n17 lt	5853
	2	1626
	2; n1 lt	5458
	2; n1 lt	767
	2 ea; n2 lt	3383
	2.0; n3 lt	2109
	2 ea; n3 lt, n1 f	1967
	2 ea; n4 lt, n1 f	8825
	2.1; r1-3; n11 lt, 1 f	4200, 11749
	2.17; r2-3; n6 lt	11035
Saguinus spp	1.63 yg; mode 2; r1-3; n295 lt	4309
Saguinus fuscicollis	1.65 yg; r1-2; n31 lt, 17 f	11874
	1.70 yg; mode 2; r1-2; n44 lt	4306
	1.72 yg; 1.44 alive; mode 2; r1-3; n43 lt, 17 f	10675
	1.82 yg; mode 2; r1-3; n148 lt	11874
	1.85 yg; mode 2; r1-3; n319 lt	3696, 3697
	1.92 yg; mode 2; r1-3; n184 lt	11874, 11875
	2.0; n5 lt, 1 f	1661
Saguinus labiatus	1.81 yg; 1.18 alive; r1-3; n27 lt	8043
	2 ea; n2 lt	3383
Saguinus midas	2 ea; n4 lt	1821, 1822
Saguinus mystax	1.5 yg; r1-2; n2 lt	4304
	1.93 yg; r1-3; n29 lt	8042
	mode 2	5584

	2; n1 lt	4306
Saguinus nigricollis	1.6; r1-2; n5 lt	4304
	1.7; r1-2; n9 lt	5767
	1.82; mode 2; r1-3; n88 lt	11874, 11875
	1.92 yg; r1-2; n13 lt	5768
Saguinus oedipus	1-2 yg; 3 rare	3503
	1.4 yg; r1-2; n5 lt	12099
	1.7 emb; r1-2; n10 f	2368
	1.72 yg; mode 2; r1-3; n138 lt	4304, 4305, 4306, 4311
	1.75 yg; mode 2; r1-2; n8 lt, 6 f	1233
	1.82 yg; mode 2; r1-3; n61 lt	7231
	1.84; mode 2; r1-3; n55 lt	10180
	1.88; mode 2; r1-3; n83 lt	3697
	1.9 emb; r1-2; n10 lt	2368
	1.91 yg; mode 2; r1-3; n68 lt	11874, 11875
	1.92 yg; r1-2; n12 lt	5768
	1.95 emb; mode 2 87.5%; r1-3; n40 lt	11852
	1.98 yg; 1.88 alive; n41 lt, 17 f	10672
	mode 2	7583
	2 ea; n2 lt	11510
	2.0; n9 lt	5767
	2.04; 1.62 alive; mode 2; r1-3; n106 lt, 47 f	10675
	2.05; mode 2; r1-3; n19 lt	3113
	2.09; mode 2; r1-3; n54 lt	5816, 5817
	2.15; r1-3	1628
	3.0; n2 lt	11733
SEXUAL MATURITY		
Callithrix jacchus	f: high progesterone: 300-400 d	2, 6, 4540, 4541, 4544, 4549
	f: 1st conceive: 335-903 d	2, 4544, 4551
	f: 1st conceive: ~14 mo; n1	6621
	f: 1st birth: 18.5 mo; r14-22; n11	8487
	f: 1st birth: 23 mo	10391
	f: 572 d; ±31.4	2, 4546, 4547
	m: high testosterone: 250-350 d	6, 4540, 4549
	m: proven fertile: ~15 mo; min 11 mo	5215
	m: testes large: min 250 d; mode 550-700 d	2, 4544, 4549, 4551
	f/m: 13 mo	8664
	f/m: 400-600 d	3
Cebuella pygmaea	f: 1st birth: 32.7 mo; median 32; sd7.6; r24-42; n7	12100a
	f/m: 2-3 yr	1822
Leontopithecus rosalia	f: 1st conceive: 29.3 mo	5856
	f: 1st birth: 2 yr	5850
	m: 1st mate: 28.3 mo; n16	5856
	f/m: 18 mo	5850
Saguinus fuscicollis	f: 1st conceive: 220 d: r211-229; n2	3053
	f: 1st conceive: 500 d; r398-631	3054
	f: 1st birth: 19-32 mo	11875
	m: 1st sire: 320 d; n2	3053
Saguinus mystax	f/m: glandular development: 18 mo	5584
Saguinus oedipus	f: 1st mate: 27.8 mo; se2.5	10672
	f: 1st birth: 24 mo; n1	7231
	f: 1st birth or abortion: 31 mo; sd8.5; n28 f	5817
	f: 1st birth: 32 mo	4306
	f: 1st birth: 34.0 mo	10672
	f: 10-11 mo; f with m; 15-17 mo; f with family	12099
	f: 436 d; ±30.2; housed with m	10671
	f: 590 d; ±54.6; housed in natal groups	10671
	m: 1st sire: 26 mo; n1	7231
	f/m: 1st mate: 15 mo	11733

CYCLE LENGTH

Callithrix jacchus	estrus: 2-3 d	6620
	follicular phase: 8.25 d; ±0.30; n56	4379, 4545
	follicular phase: 8.8 d; ±3.7	4372
	follicular phase: 9.5 d; sd2.2; r6-12; n10	5719
	luteal phase: 16.6 d; sd1.1; r15-18; n9	5719
	luteal phase: 19.22 d; ±0.63	4379, 4545
	luteal phase: 21.5 d; ±2.2	4372
	mode 24-30 d; r9-32; n14 cycles, 8 f	3
	16.4 d; se1.7; n9 cycles	4549, 4550
	26.2 d; sd2.4; r24-29; n8 cycles, 5 f	5719
	27.5 d; n40 cycles, 14 f	4379, 4546, 4547
	28.54 d; se0.95; n18 cycles, 18 f with intact m	4378
	28.63 d; se1.01; n19 cycles, 14 f isolate or with castrate	4378, 4544, 4545
	30.1 d; sd3.8; median 29.5; r23-41; n30 cycles, 15 f	4372
Cebuella pygmaea	estrus: 6 d	10199
Leontopithecus rosalia	~15 d	3509
	2-3 wk	5850, 5851
	19.56 d; se1.41; n14 cycles, 4 f	3508
Saguinus fuscicollis	14.9 d; se0.7; r13.5-21.8; n18 cycles, 4 f with m?	3054
	17.3 d; se0.97; r13.5-21.8; n34 cycles, 6 f in family group	3054
	17.5 d; se1.0; r14.6-18.2 80%; r13.6-21.1; n10 f	3055
Saguinus oedipus	15.5 d; ±1.5; n15 cycles, 5 f; between progesterone peaks	8748
	21-32 d; between LH peaks	12099
	22.7 d; se1.7; r19-25; n10 cycles, 4 f	1234
	23.6 d; se1.2; n22 cycles, 5 f	3504

GESTATION

Callithrix jacchus	implant: by 10 d	4542
	implant: 10-12 d	4378, 4545
	implant: 11-12.5 d	7497
	~132 d	7384
	133 d; r117-156; n3	6749, 6751
	138 d; r126-150; n2 lt, 1 f	1753
	est 140 d; r~138-170; n6	6620, 6621
	est 140-150 d	3049
	142 d	4567
	143.4 d; r141-146; n10	10388
	143.5; ±2.5	8253
	median est 144 d; n129	8663
	144 d; sd2; r141-146; n9	1686, 4544, 4545, 4546, 4547, 4549
	144-146 d	9314
	mode 144-185 d; r144-402; n28; pairing to birth	5952a
	145.3 d; r143-147; n3; pairing to birth	4307, 4308
	146 d; n1	9313
	min 146 d; 166 median; pairing to birth	1167
	21 wk	6647
	148 d; se4.3	4540, 4550
	151 d; r150-153; n4	4082, 8488
Cebuella pygmaea	107 d; min ibi	6222
	19.1 wk; r17-20; n5	1821, 1822
Leontopithecus rosalia	122 d	3508
	128.6 d; r125-132; n9	5850, 5851
	max 134 d	8825
	135 d	2109
	max 136.8 d; ibi	11749
Saguinus bicolor	~160 d	4591
Saguinus fuscicollis	145-152 d; n1	11874
Saguinus mystax	150 d	5584
Saguinus oedipus	>4.5 mo; n4; signs of pregnancy to birth	11594
	144-150 d	11510
	147 d; n1; pairing to birth	4304, 4305, 4306
	165-170 d	11733
	165.5 d	2153
	166-202 d; n1; pairing to birth	1233

	170-172 d; n1	1235
	183.7 d; se1.14; r179-187; n6 f; LH peak to birth	12098
	305 d; sd71.9; n5; pairing to birth	3118
LACTATION		
Callithrix humeralifer	nurse: 3-4 mo	9401
Callithrix jacchus	solid food: 23 d	10391
	solid food: 25 d	4550
	solid food: 4 wk	5147
	solid food: 4 wk	6813
	solid food: 4 wk; n1	6620
	solid food: 30.2 d; r26-34; n6	3049
	wean: 35 d; n1	6751
	nurse: 5-6 wk; n1	3289
	nurse: mode 40 d; max 80	4540, 4551, 4550
	wean: 40-100 d	4546
	wean: 6 wk; n1	6620
	wean: r52-85 d; n5; r52-60; n4	3049
	wean: 2 mo	1753
	wean: 60-81 d	1822
	wean: 80-100 d	6648
	nurse: 129 d; n1	1166a
Cebuella pygmaea	suckling decreases: 3 mo; n1	6222
	suckle: 3 mo	1821, 1822
	independent: 5 mo	1822
Leontopithecus rosalia	solid food: 4 wk	5854
	solid food: 5 wk	215
	nurse: 10 wk	215
	wean: 12 wk	5854
Saguinus oedipus	solid food: 25 d; n1 lt	11510
	solid food: 34.5 d; r29-40; n2	3049
	wean: 50 d	3503
INTERLITTER INTERVAL		
Callithrix argentata	median 6.5 mo; r5-13	10390
	median 7.5 mo; r5.5-10	10390
Callithrix jacchus	4.5-5.0 mo	4545
	most 147-167 d; r147-290; n166	5952a
	148 d; n1	4306
	5 mo	10391
	5-6 mo	4541
	153.6 d; r151-156; n9, excluding one at 275 d	10388
	mode 154 d; r145-190; n>60	8663
	~155 d	4540
	155.5 d; r155-156; n2	9313
	median 158 d; r145-382; n>130	1167
	159 d; n1	6749, 6751
	159.8 d; r150-267; n28 lt, 5 f	9314
	160-200 d	4550
	160-262 d; n3 f	3049
	171 d; r152-305; n28	8487
	179.1 d; sd28.1; median 175; r151-246; n22 after abortion	6648
	188 d; r126-335; n10	3697
	204.9 d; sd55.2; median 186.5; r147-441; n152 (n38 yg died)	6648
	212.6 d; sd57.7; median 206.0; r147-441; n114 yg lives	6648
Cebuella pygmaea	158.96 d; r148-164.8; n8 f; r107-191; n32 intervals	6222
	19-20 wk	1822
	5-7 mo	10198
	21-23 wk	1821
	185 d; n2 f	4309
	212.7 d; sd122.3; median 166; r149-746; n31 intervals	12100
Leontopithecus rosalia	132.5 d; r132-133; n2 minimum lengths	11035
	8.1 mo; r4-19; n11 lt, 10 intervals	4200
Saguinus fuscicollis	~6 mo	1661
	222 d; r162-367; n60	3697
Saguinus labiatus	224 d; r165-317	8043

Saguinus mystax	150 d	5584
	187 d; 1st to 2nd birth; 274 d; 2nd to 3rd; 201 d; 3rd to 4th	8042
Saguinus oedipus	172 d shortest	1628
	207 d; sd16.9; r195-238; n5 yg die	3113
	208.5 d; sd16.3; n35 intervals, 8 f	3503
	222.7 d; r154-265; n7	7231
	238 d; r187-302; n13 full term	4304
	245 d; r189-334; n8 aborted	4304
	259.1 d; r166-523; n17 hand reared	11874
	280.0 d; r191-578; n17 parent reared	11874
	7-8 mo	11594
	289 d; median 238; sd16.9; r182-374; n7 yg live	3113
	294 d; se18.2; r160-480; n24	5816, 5817
	295 d; r190-446; n27	3697
SEASONALITY		
Callithrix argentata	birth: yr rd; n41; zoo	1232
Callithrix humeralifer	mate: est Sept-Nov, Feb-Apr; Brazil	9401
Callithrix jacchus	mate: May, Feb, July; n1 pair; zoo	6749
	conceive: low July-Aug; captive	12102
	birth: Feb; few data; captive	9313
	birth: end Oct-Nov, end Mar-Apr; Brazil	5059
	birth: peak spring, fall; zoo	1230
	birth: yr rd; zoo	9314, 9315
	birth: yr rd; lab	4550
	birth: yr rd; n92; captive	8487
	birth: yr rd; n30; zoo	1232
	repro: yr rd; captive	1167
Cebuella pygmaea	birth: July; ne Peru	8891
	birth: peaks Nov-Jan, May-June; Peru	10198
Leontopithecus rosalia	birth: Feb, Apr, July; n1 pr; zoo	2109
	birth: peak Feb-May, July-Aug; captive	5850
	birth: Sept-Feb; Brazil	5854
	birth: Sept-Mar; Rio de Janeiro, São Paulo, Brazil	1972
	birth: yr rd; captive	10185
Saguinus fuscicollis	birth: peak Feb-Mar; lab	3696
	birth: yr rd; n40; zoo	1232
	birth: yr rd, peak Mar-June; lab	3697
Saguinus mystax	birth: middle & end yr; captive	5584
Saguinus oedipus	mate: peak Nov-Dec; captive	2153
	ovulate: Nov-Feb; Panama	2366
	preg f: Feb-July; Barro Colorado I, Panama	3024
	birth: Mar; only 3 mo data; Carribbean Atlantic Coast	6477
	birth: Mar-June; Panama	2366
	birth: peak Mar-Jun, low July-Sept; n44; zoo	1232
	birth: small peak Apr-May, none Sept; captive	2153
	birth: yr rd, peak Mar-May; lab	5816, 5817
	birth: yr rd, peak end Apr-early June; Panama	2368
	birth: yr rd, peak Nov-Mar 28/46, none Aug; captive	4304
	birth: yr rd; zoo	7231
	young seen: Feb-early June, peak May; n Colombia, central Panama	7583

Order Primates, Family Callimiconidae, Genus *Callimico*

Goeldi's marmoset (*Callimico goeldii* 450-850 g) is an arboreal omnivore from the Amazon basin. Placentophagia is reported (698, 8651). Estrus occurs 1-10 d postpartum (698, 2153, 4613, 6577, 6923, 12097). Neonates are furred with open eyes and suckle from axillary teats (698, 4580, 4613, 4792, 8651). Males and siblings carry preweaned young (4612, 4613, 8651, 8653). Details of endocrinology (1629, 1630, 1823, 12101) are available.

NEONATAL MASS		
Callimico goeldii	est 25-30 g	6579
	44 g	8651
	m: 52.3 g; r45-66; n3	698
	f: 54.7 g; r46-63; n3	698
	m: 59.8g; n1	4733
NEONATAL SIZE		
Callimico goeldii	f: 91.0 cm; n1; 4 d	4792
LITTER SIZE		
Callimico goeldii	1	698
	1	7584
	1 ea; n2 lt	8651
	1 yg ea; n7 lt, 3 f	4580
	1 yg ea; n7 lt	6579
	1 yg ea; n9 lt	8653
	1 yg ea; n11 lt	4306
	1 yg ea; n>20 lt	4613
	1 yg ea; n25 lt	4309, 4313
	1.02; r1-2; n188 lt	214
	2; n1 lt	4792
SEXUAL MATURITY		
Callimico goeldii	f: 1st conceive: min 12 mo	698
	f: 1st birth: 16.5 mo; n1	6577
	m: 1st mate: min 13 mo	698
CYCLE LENGTH		
Callimico goeldii	estrus: 7 d; r5-12	6577
	22 d; r21-24; n13 cycles, 3 f	6577
	24 d; few data	698
GESTATION		
Callimico goeldii	≤143, 144, 156 d; n3	8653
	143.75 d; r143.5-144; n2; pairing to birth	8651
	150-160 d; ibi	2153
	150-180 d	6923
	154 d; r151-159; n7 lt, 3 f	6577
	160 d; r154-167	698
LACTATION		
Callimico goeldii	solid food: 18-27 d; n3	6923
	independent: 33-40 d; n3	6923
	wean: max 10 wk	4580
	wean: 12 wk	4613
INTERBIRTH INTERVAL		
Callimico goeldii	165 d without extremes; r161-290	698
	167 d; n5 births, 1 f	6579
	187 d; n3 f	4309
	197.7 d; r165-245; n6, 3 f	8653
SEASONALITY		
Callimico goeldii	birth: Mar-Dec, none Jan-Feb; zoo	2153
	birth: Nov; n1; Bolivia	8654, 8655
	birth: yr rd; captive	698
	birth: yr rd; n37; zoo	1232

Order Primates, Family Cebidae

New World monkeys (*Alouatta* 4-10 kg; *Aotus* 600-1200 g; *Ateles* 3.5-8 kg; *Brachyteles* 9.5 kg; *Cacajao* 3.5-11 kg; *Callicebus* 500-1000 g; *Cebus* 1-4.5 kg; *Chiropotes* 2.7-3.2 kg; *Lagothrix* 4-11 kg; *Pithecia* 700-2500 g; *Saimiri* 500-1500 g) are arboreal, often social omnivores from the forests of Central and South America. Males are often larger than females (1409, 1609, 1974, 3382, 5110, 5394, 5395, 10807, 10808; cf 350, 11189, 11940).

A copulatory vaginal plug is reported (*Cebus* 4296). The hemochorial, labyrinthine, villous, discoid or bidiscoidal placenta, which secretes gonadotrophic hormones (10636a), has an antimesometrial, superficial implantation into the dorsal and ventral wall of the simplex uterus (4113, 4788, 4790, 7563, 10310, 10415,

10475, 11837, 11843, 11845, 11855, 11966). Placentophagia (1154, 1609, 1840, 4791, 4974, 5520, 9743, 10636, 11940), but not estrus (2622, 3800, 5100) occurs after the birth of an open-eyed and furred neonate (399, 2622, 3043, 3449, 4791, 7225, 7296, 9286). *Aotus* and *Callicebus* males contribute to parental care (3449, 5770, 6932, 11577, 11941). Seminal vesicles and a baculum (except in *Aotus*, *Ateles*, and *Lagothrix*) are present (4788, 5842).

Endocrinological data for squirrel monkeys (*Saimiri*) are not extensively cited here as these monkeys are common laboratory models for human endocrinology. Details of anatomy (1894, 1895, 2477, 4685, 4696, 4788, 5495, 11847, 11851, 11865), endocrinology (*Aotus* 1094, 1095, 1097, 2622, 2628, 2629, 4235, 5100, 8781, 9298, 9795, 10793, 11865; *Ateles* 4866; *Cebus* 1656, 2201, 4863, 4866, 6229, 7741, 7742, 7743; *Lagothrix* 1656), birth (1154), and milk composition (1481a, 1840) are available.

NEONATAL MASS		
Alouatta palliata	275-400 g	3800
Aotus trivirgatus	56 g; n1	5904
	90-105 g	2621
	91.92 g; sd21.04; n13	2982
	96.5 g; r95-105; n4	2622
	m: 98 g; n1	3043
Ateles fusciceps	<500 g	2931
Ateles geoffroyi	340 g; n1	7296
Ateles paniscus	66.7 g; r60-80; 3 stillborns, est premature	1558
	480 g, n1	3710
Callicebus moloch	74.4 g; n1	5390
	~100 g	3449
Cebus albifrons	f: 222.1 g; sd9.1; n14	3295
	226 g; se5.8; r184-244; n10	11694
	m: 250.7 g; sd18.9; n20	3295
Lagothrix lagotricha	f: 140.5 g; n1, 2 d	11717
Saimiri sciureus	62-247 g; high mortality <80 g	8938
	87 g; r68-100; n4	1154
	84.3 g; r78-90; n3 dead	4974
	80-140 g	536
	90-125 g	2790
	90-125 g	2042
	f: 93 g; n1	10889
	94.5 g; r84-113; n6	5113
	f: 100 g; min 80	10451
	f: 104.1 g; se1.3; n193	8938
	104.85 g; r73-115; n20	3963
	109.49 g; sd13.4	11406
	m: 111.4 g; se1.1; n254	8938
	120.9 g; n10?	5686
	129.1 g; sd7.14; n4	5475
NEONATAL SIZE		
Aotus trivirgatus	CR: 12.79 cm; sd1.99; n13	2982
Callicebus moloch	HB: 125 mm; n1	5390
Cebus apella, C albifrons	CR: 14.7 cm; sd0.5; n7	460
Saimiri sciureus	~12-12.5 cm	5475
	CR: 12.26 cm; sd0.54	11406
	12.6 cm; n1	4791
	CR: 12.9 cm; sd0.4; n8	460
	13.2 cm; n4	5475
WEANING MASS		
Saimiri sciureus	f: 403 g; n1, 177 d	10889
LITTER SIZE		
Alouatta caraya	1 yg ea; n7 lt	9921, 9922
Alouatta palliata	1	6387
	1; n1 lt	7624
Alouatta pigra	mode 1	3799
Alouatta seniculus	1	5110
	1	5070
	1 yg ea; n3 lt	3383

	2; n1 lt	2147
Aotus trivirgatus	1	11941
	1	2621
	1 yg; n1 lt	5904
	1; n1 lt	3043
	mode 1	3024
	1 yg; 0.5% twins; n179	3985a
	1.06; r1-2; n35 lt	1840
	2	11869
Ateles geoffroyi	1; n1 lt	7296
Ateles paniscus	1	11188
	1 emb; n1 f	5110, 5111
	1 yg ea; n5 lt	10021
Cacajao calvus	1 yg; n17	3380, 3382, 3383
Callicebus moloch	1	11941
Callicebus torquatus	1	5770
Cebus albifrons	1 yg	2429
Cebus apella	1 ea; n7 lt	1230
	1 yg; n19+ lt	10468
	2; n1 lt	2943
	2 yg; n1 lt	10468
Cebus capucinus	1; n1 lt	11086
Chiropotes albinasus	mode 1	10809
Chiropotes satanas	1	11189
	1 yg; n2 lt, 1 f	4722
Pithecia pithecia	1	9923
	1 ea; n2 lt	1893
	1 ea; n2 lt	4722
Saimiri sciureus	1	4791
	1 yg	7117
	1 ea; n21 lt	3650
	1.002; r1-2; n512 lt	8938
SEXUAL MATURITY		
Alouatta caraya	f: 1st conceive: 38.5 mo; r35-42; n2	9922
	m: 1st sire: 30.5 mo; r24-37; n2; perhaps earlier	9922
Alouatta palliata	f: 1st estrus: 4-5 yr	1609
	f: 1st birth: 3-5 yr	3531
	f: 1st birth: 43 mo	3800
	m: 30-48 mo	3800
	m: 6-8 yr	1609
Aotus trivirgatus	f: 1st birth: 40.56 mo; sd7.82; n9	3985a
	f: 1.5-2.0 yr	4547
	m: high testosterone: median 313 d; r211-337; n6	2622, 2629
	m: 1st sire: 42.17 mo; sd10.73; n12	3985a
	f/m: est 18-24 mo	2621
Ateles geoffroyi	f: 1st birth: 5.7, ~6.5, 7.6 yr; n3	7355
	f: 5 yr	2929
Cacajao calvus	f: 1st birth: 3.58 yr; n1	3382
	m: 1st mate: 17.5 mo; r17-18; n2	3382
	m: 1st sire: ~6 yr	3380, 3382
Cebus albifrons	f/m: 3.59 yr; se0.17; r2.83-4.4	11694
Cebus capucinus	f: physically mature: 4 yr	8117
	m: physically mature: 8 yr	8117
Lagothrix flavicauda	f/m: > 4 yr	6643
Lagothrix lagotricha	f: 1st birth: 8 yr; r6-10; n2	11717
	m: sexual interest: 3-4 yr	11717
Pithecia pithecia	m: > 4 yr	4722
Saimiri sciureus	f: 1st conceive: 35.03 mo; sd5.18; r32.11-42.78	9354
	f: 1st conceive: 58 mo; r44-67; n6	5520
	f: 1st preg: mode 3.5-4 yr n24/60; r2.5-8.5	10687
	f: 1st birth: 40.57 mo; sd5.18; r37.66-48.33	9354
	f: 3 yr	4547
	m: 1st sire: 42 mo	9354

CYCLE LENGTH

Alouatta palliata	estrus: several d	1606
	15.52 d; sd4.88; n25 cycles, 14 f	5395
	16.08 d; sd4.27; n12 cycles, 10 f	5395
	16.3 d; se0.7; r11-24; n23 cycles, 6 f	3800
Alouatta seniculus	estrus: 2.5, 3 d; n2	2149
	20.5 d; median 17; r16-27 d; n4 cycles, 4 f	2149
Aotus trivirgatus	follicular phase: ~6 d	2622
	luteal phase: ~10 d	2622
	10-15 d; n6 f	1840
	15.92 d; se0.26; n60 cycles, 55 f	1094, 1095, 1097, 2621, 2622
Ateles fusciceps	estrus: 2 d	2929
Ateles geoffroyi	menses: 3-4 d; n1 f	3895
	24-27 d	3895
Cacajao calvus	14-48 d	3381
Cebus spp	menses: 4.3 d; r1-7	4296
	18.3 d; mode 18; r16-23 d; n12 cycles, 2 f; vaginal smears	4296
Cebus albifrons	menses: 2-4 d	1656
	17-19 d	4866
	17.5 d; r15-21; n50 cycles	1656
Cebus apella	estrus: 5-6 d	5254
	menses: 2.8 d; ±0.4; r1-5	7741
	menses: 4.06 d; sd1.89; r0-8; n18	11937
	17.08 d between menses; ±4.27; r11-24; n13; 21.14 d with menses	11937
	20.8 d; mode 21; r13-28; n108 cycles	7741
Cebus griseus?	median 14-16 d; r10-19	3770
Lagothrix lagothricha	estrus: 3-4 d	11718
	menses: 3 d	11717
	r12-49 d; n25 cycles, 2 f	1656
	3 wk	11717
Saimiri boliviensis?	9.7 d; r6-12	148a
Saimiri sciureus	estrus: 12-36 h	9287
	6-12 d	2541
	7; mode 7; n15	9287
	mode 8; n42 cycles, 9 f	11776
	8.7 d; sd0.6; r8-10; between progesterone peaks	11868
	9.4 d; se0.6; r8-10; between estrogen peaks	11868
	9.44; r6-12; n9 cycles, 7 f	2541
	9.7 d; se0.4; n20 cycles, 10 f	12012
	10.9; se0.10; r9-13; n9 cycles, 6 f	10889
	11.3 d; n10 f; 8.2 d; 10 f smaller cage	5113
	12 d; sd2.7; median 10-11; n108 cycles, 24 f	1656
	12.54 d; ±2.01; mode 13; r6-18; n23 f	6176
	13.36 d; r8.3-25.3; n53 cycles, 5 f	5113
	18-20 d; n15 f	10287

GESTATION

Alouatta spp	139 d	11869
Alouatta caraya	187 d; n1	9921, 9922
Alouatta palliata	186 d; r180-194; n4; swelling to birth	3800
Alouatta pigra	180-194 d	2932
Alouatta seniculus	191 d; r186-194; n6	2149, 9743
Aotus trivirgatus	120-140 d; palpation, ibi	1840
	18 wk; n17 f	4235
	135.5 d; r133-138; n2	5100
	est 148-159 d	2982
	~5 mo; min ibi	7224
Ateles belzebuth	139 d; n1	1558
Ateles fusciceps	7-7.5 mo; max 226 d	2929
	226-232 d	2931
Ateles geoffroyi	~200 d; n1	7355
Callicebus spp	5-6 mo	7225
Cebus spp	162 d; r157-167	11869
Cebus apella	152-160 d	2943

	153 d; r149-158; n3	11937
Lagothrix lagothricha	7 mo	11718
	225 d; n1	6696
	7.5 mo	11717
Pithecia pithecia	*est* 101 d; n1	4316
	est ~150 d	9923
	163-176 d; n1	4722
	169.5; r163-176; n2	4722
Saimiri sciureus	146.9 d; sd3.28; median 146.5; mode 146; r141-154; n10	5686
	152.5 d; r141-157; n46	10451
	157 d; n1	10889
	159 d; se3	1974
	162.6 d; r152-168; n5 f	6578
	24-26 wk; n13	4974
	24-26 wk	3650
	170.8 d; r168-172; n19	3963
	177.75 d; sd7.2; n4	5475
	~6 mo	4791
	6 mo; field data	2791
LACTATION		
Alouatta caraya	nurse: min 8 wk	9921
Alouatta palliata	wean: min 6 mo	1609
Alouatta seniculus	wean: 10.5-14 mo	2147
	wean: >1 yr	6695
Aotus trivirgatus	solid food: 19 d	7224
	solid food: 2 mo	399
	wean: 8-10 wk	7224
	wean: 12-14 wk	7225
	wean: ~9 mo	11940
Ateles fusciceps	solid food: 8 mo	2931
	wean: min 16 mo	2931
	wean: ~10-12 mo; n6 yg	2929
Ateles geoffroyi	wean: 2+ yr; 26-28 mo	7355
Cacajao spp	wean: ~21 mo	3381
Callicebus spp	wean: 12-16 wk	7225
Callicebus torquatus	wean: > 4 mo	5770
Cebus albifrons	wean: 9 mo	11694
Lagothrix lagothricha	nurse: *est* 9-12 mo	11717
Pithecia pithecia	nurse: 4 mo	4316
Saimiri sciureus	solid food: 3 mo	9286
	wean: 177 d; n1	10889
INTERBIRTH INTERVAL		
Alouatta caraya	11.1 mo; r7-16; n16	9922
Alouatta palliata	22.5 mo; se0.6; n16	1886, 3800
Alouatta seniculus	10.5 mo; sd2.9; r7.75-15; n12 yg dies	2150
	17.4 mo; sd4.5; r10.5-35; n135 yg lives	2148
Aotus trivirgatus	204, 206, 274 d; n3	399
	213.5 d; r156-319; n9	7224
	253 d; r166-419	2621
	271 d; median 258; min 148; n51	5100
	1 yr	11940
	12.72 mo; sd5.72; r5-29; n75 intervals	3985a
Ateles fusciceps	22-36 mo	2931
Ateles geoffroyi	31.9 mo; ±3; r28-36; n7 birth, 3 f	7355
Ateles paniscus	34.5 mo; sd5.8; r25-42; n17 intervals	10609
	3 yr	11188
Brachyteles arachnoides	*est* >24 mo	7356
Cacajao calvus	23-28 mo; n5	3380, 3382
Callicebus moloch	1 yr	11941
Pithecia pithecia	10-11 mo	11739
Saimiri sciureus	1 yr	5520

SEASONALITY

Alouatta caraya	repro: not seasonal; zoo	9922
Alouatta palliata	sex behavior: not seasonal; Barro Colorado I Panama	1606
	all stages of embryonic development: July-Aug (2 yr); Panama	11845
	birth: Feb, Apr-June, Aug-Sept; n16; Guanacaste, Costa Rica	5394
	birth: yr rd; Guanacaste, Costa Rica	1886, 3800
Alouatta pigra	young of various age seen in most bands; Veracruz, Mexico	4228
Alouatta seniculus	birth: yr rd, peak end dry, low early wet season; Guarico, Venezuela	2147
Aotus trivirgatus	mate: Dec; n1; Barro Colorado I, Panama	3024
	birth: Jan, Mar, May, Aug, Oct-Nov; n9; zoo	7224
	birth: June; Barro Colorado I, Panama	3024
	birth: yr rd, peak Oct-Jan; n126; captive	3985a
	repro: yr rd; captive	2622, 5100
Ateles belzebuth/A geoffroyi	birth: yr rd; n15; zoo	5862
Ateles fusciceps	birth: Apr-June; n11/19; captive	2931
Ateles geoffroyi	all stages emb development: July-Aug; Panama	11845
	birth: Feb-Mar; Chiapas, Mexico	218
	birth: June-July; Nicaragua	11837
	birth: peak June-Dec; Barro Colorado I, Panama	7355
	birth: yr rd; Yucatan, Mexico	3633
	repro: yr rd; Panama	1607
Ateles paniscus	birth: peak Jan-Aug; Peru	10608
	young seen: yr rd; Suriname	5110
Cacajao calvus	mate: Oct-May; zoos, n hemisphere	3381, 3382
	birth: May-Oct; zoos, n hemisphere	3381, 3382
Callicebus moloch	birth: Nov-Mar; equatorial S America	5770, 6932
Callicebus torquatus	birth: Nov-Mar; Peru	5770
Cebus spp	ovulate: yr rd; s Goyaz, Brazil	4296
	birth: peaks May-June, Oct-Nov; s Goyaz, Brazil	4296
Cebus albifrons	conceive: Dec-Jan; Peru	1022
Cebus apella	estrus: yr rd, peak Apr-June; no data Jan-Feb; Peru	5254
	birth: mid-end Jan, early-mid May, Oct-mid Dec; zoo	1230
	birth: peak Oct-Dec; Peru	5254
	young seen: yr rd; Suriname	5110
Cebus capucinus	all stages of embryonic development: July-Aug; Panama	11845
	birth: 3/5 peak Jan, Mar-Apr; Barro Colorado I, Panama	8117
	birth: dry season, Dec-early Apr; Costa Rica	3501
Chiropotes satanas	birth: Dec-Jan; Guyana	11189
Lagothrix lagothricha	repro: not seasonal; captive	11718
Saimiri sciureus	testes: large Apr-Aug; captive, Argentina	1974
	vaginal cornification: high Jan-Aug, low Oct; lab	9287
	estrus: yr rd, peak June-Sept; lab	10889
	mate: June-Aug; captive, Argentina	1974
	mate: July-Sept; Amazon, S America	2790
	mate: Aug-Feb; lab	11776
	mate: Sept-Mar; zoo	2790
	mate: after Oct; lab	12012
	mate: Nov-Mar; captive	7202
	mate: Dec-Mar; lab	2542
	mate: mid Dec-Mar; captive	2791
	conceive: mid Dec-mid Mar; lab	11719
	preg f: Oct-Jan; captive	11869
	preg f: yr rd; captive	9288
	birth: Jan-Mar; Amazon, S America	2790
	birth: Feb-Sept; zoo	2790
	birth: May-July; captive	9354
	birth: June-Aug; captive	2791
	birth: June-Aug; lab	2542
	birth: 65% June-Aug, 87% May-Sept; lab	5520
	birth: peak Oct-Nov; captive, Argentina	1974
	birth: end July-Oct; Surinam and captive	3650
	birth: yr rd, 72% May-Aug; captive	8938

Order Primates, Family Cercopithecidae

Old World monkeys (*Allenopithecus* 3.5-6 kg; *Cercocebus* 3-18 kg; *Cercopithecus* 750 g-12 kg; *Colobus* 3-23 kg; *Erythrocebus* 4-25 kg; *Macaca* 1.5-20 kg; *Nasalis* 7-25 kg; *Papio* 6-54 kg; *Presbytis* 5-21 kg; *Pygathrix*, *Theropithecus* 10-20 kg) are primarily arboreal, social herbivores or omnivores in Africa and Asia. Males are often larger than females (92, 146, 790, 1015, 1116, 1297, 1781, 1952, 2501, 2955, 5692, 5703, 5758, 6970, 7270, 8496, 9667, 9970, 9982, 10078; cf 1115).

The 20-290 g, hemochorial, villous, bidiscoid placenta, which secretes gonadotrophins (10636a), has a superficial, antimesometrial implantation in the simplex uterus (679, 1787, 1903, 2580, 3748, 4113, 4721, 4723, 4786, 4796, 4842, 5009, 5010, 5011, 5228, 5316, 6624, 7563, 7946, 8172, 8174, 8222, 9707, 9728, 10209, 10310, 10311, 11199, 11843, 11852, 11861, 12129) and may be eaten after birth (16, 49, 1798, 1903, 2501, 3762, 3966, 4298, 4445, 4448, 5273, 5649, 5919, 6606, 7768, 7804, 7946, 9289, 9334, 9982, 10142, 10234, 10264, 10636, 10726, 10914, 11199; cf 1299, 4720, 4723, 7485). Estrus postpartum is uncommon, but has been observed in *Macaca fascicularis* (3385, 10844). Sex of offspring may be related to the social rank of the mother (8265, 8265a; cf 10162a). Neonates have fur and open eyes (184, 482, 1042, 1798, 2495, 3386, 3966, 4721, 5273, 5758, 6747, 8118, 8701, 8705, 9361, 9728, 10158, 11985). Females have 2 mammae (331, 1483, 10158). Males have seminal vesicles (2293).

Endocrinological data for macaques (*Macaca*) and baboons (*Papio*) are not cited as these data are summarized in reviews of human endocrinology (5891). Details of anatomy (3384, 5508, 12125), endocrinology (*Cercocebus* 91, 92, 1553a, 6770, 6771, 10291; *Cercopithecus* 256, 689, 1165, 1204, 2631, 2888, 2889, 2967, 2988, 2989, 4699, 4860, 5703, 6081, 6874, 7187, 9334, 9701, 9794, 10368a; *Erythrocebus* 92, 456, 11826; *Presbytis* 4786, 6547, 8884, 9832, 9833; *Pygathrix* 2201, 6229), copulation (3642), and birth (16, 1795a, 1798, 2800, 3892, 3966, 4298, 4616, 6606, 7083, 7507a, 8118, 8503, 8974) are available.

NEONATAL MASS

Cercopithecus aethiops	est 300-400 g	10154
	f: 310 g; r220-419; n26	11062
	m: 319 g; r219-411; n28	11062
	321.3 g; sd56.9; n38 (14 f, 24 m)	5364
	m: 352 g; r233-480; n26	9728
	f: 364 g; r247-480; n24	9728
	397 g; n1	6612
Cercopithecus ascanius	371 g; term	3748
Cercopithecus mitis	337 g	3748
	360 g; n1	9362
	402.2 g	8107
Cercopithecus talapoin	~170 g	3640
	230 g; n1	9338
Cercopithecus wolfi	435 g	3748
Colobus polykomos	m: 820 g; n1	2109
Macaca arctoides	450 g	11985
	495 g; n1	10914
	500 g; r380-650; n38	7908
Macaca cyclopis	402 g; sd61; n54	8347
	402 g; r300-530; n35	12000
Macaca fascicularis	272 g; sd50; r210-340; n≤11	11549
	m: 287 g; n1	10263
	~300 g	3550
	f: 302.2 g; sd49.5; r240-360; n9; gang cages	4954
	m: 302.7 g; sd52.6; r240-370; n5; gang cages	4954
	m: 309.7 g; sd38.0; r240-410; n41; individual cages	4954
	f: 320.0 g; sd48.6; r200-450; n46; individual cages	4954
	f: 335 g; r234-469; n83	11062, 11064
	338 g; sd80; r153-470; n24	814
	m: 359 g; r242-546; n81	11062, 11064
Macaca fuscata	475 g; r450-500; n2	4524
	f: 517 g; sd90; n4	2291
Macaca mulatta	412, 450 g; n2	4445
	421 g; sd78; n271 live, 390 dead; wild bred	11061

	f: 424 g; r385-475; n5	386
	f: 435 g; ±51; r351-570; n20	6880
	~442 g; r320-500; se49.3; n97	5228
	f: 450 g; sd12; n303	10077, 10078
	453 g; r340-660; n22	4448
	f: 456.2 g; sd69.9; n88	3545
	f: 457.7 g; r260-670; n161	11199
	m: 467.5 g; r395-540; n2	386
	f: 469 g; ±77; r247-752; n193	6880
	m: 470 g; sd70.5; n91	3545
	472 g; sd56; n27	5693
	472 g; sd65; n111, 160-169 d gestation	11201
	f: 474 g; sd74; n229 live, 23 dead	11061
	f: 477 g; sd77; n285	11064
	m: 479.1 g; r300-770; n150	11199
	m: 480 g; sd13; n262	10077, 10078
	480.5 g; sd78.2; n209	2580
	f: 482 g; sd52; r424-550; n4	4149
	f: 484 g; r340-678; n69	11062
	489 g; sd49; n469 live, 61 dead, 22 dead unsexed	11061
	m: 494 g; ±73; r377-756; n40	6880
	498 g; sd70.7; n100, 170-179 d gestation	11201
	m: 500 g; sd75; n276	11064
	m: 501 g; ±68; r320-760; n222	6880
	m: 504 g; r362-806; n77	11062
	m: 513.2 g; sd9.85; n67	1297
	m: 516 g; sd74; n5	4150
	536 g; sd85.5; n19, 180-189 d gestation	11201
	544.4 g; sd101.6; n8, 175 d gestation	5692a
Macaca nemestrina	400-600 g; 176 d gest	7865
	f: 451 g; sd70; n187	10009
	m: 493 g; sd70; n211	10009
	500-600 g; nll; from graph	10304
Macaca radiata	290 g; n1 stillborn	4450
	350 g; r330-370; n2	4450
	f: 388 g; sd34; n5	11062, 11064
	m: 411 g; sd50; n11	11062, 11064
Macaca silenus	m: 348.4 g; n1	2109
	f: 430 g; n1	5650
Macaca thibetana	~550 g	3386
Nasalis larvatus	454 g; n1	9667
Papio hamadryas	m: 600 g; n1 near term	4796
	600-800 g	3783
	f: 820.2 g; n15	7095
	m: 910.4 g; n9	7095
	f: 910 or 920 g; sd120; r0.70-1.40; n66	1952, 1953, 3807, 3808, 3809
	861 g; r590-1212; n17	3776
	m: 980 g; sd130; r0.67-1.22; n77	1952, 1953, 3807, 3808, 3809
Presbytis entellus	500 g; n1 term emb	4786
Presbytis obscura	113.3 g; n2	482
Presbytis senex	360 g; n1	4785
	490 g; r390-590; n2	4783
Pygathrix nemaeus	80 g; premature	4721
NEONATAL SIZE		
Macaca mulatta	CR: 16.6 cm; sd1.0; n5	460
	18.5 cm; se1.43; n12	5228
	30.5 cm; sd1.2; n8	5693
Macaca nemestrina	CR: r19.0-21.5 cm; 176 d gest	7865
Papio hamadryas	19 cm; near term	4796
	f: CR: 23.9 cm; sd1.1; r21.5-29.4; n59	1952, 3807
	m: CR: 24.1 cm; sd1.1; r21.0-27.0; n73	1952, 3807

WEANING MASS

Cercopithecus aethiops	m: 1015 g; r700-1300	9728
	f: 1058 g; r830-1450	9728
Macaca fascicularis	f: 686 g; sd206; r535-988	11595
	m: 844 g; sd76; r728-963	11595
	858 g; n28	11595
	m: 860 g; sd86; r780-930	11595
	f: 994 g; sd166; r759-1185	11595
Macaca fuscata	805 g; r780-830; n2, 4 mo, small	7758
Papio hamadryas	f: 3840 g; sd1060; r1000-8300; n241, 1 yr	1952
	m: 4060 g; sd1.15; r1600-8200; n247, 1 yr	1952

WEANING SIZE

Papio hamadryas	f: CR: 39.6 cm; sd3.3; r31.7-51.7; n242, 1 yr	1952
	m: CR: 40.0 cm; sd3.8; r28.4-51.6; n249, 1 yr	1952

LITTER SIZE

Allenopithecus nigroviridis	1 yg ea; n3 lt, 1 f	8705
Cercocebus albigena	1 ea; n5 lt	2109
Cercocebus galeritus	1 yg; n1 f	5768a
Cercocebus torquatus	1 ea; n5 lt	1230
Cercopithecus aethiops	1; 2 rare	9945
	mode 1; r1-2	2449
	1; n1 lt	187
	1 emb; n1 lt	4926
	1; n4 lt, 1+ wk	331
	1 ea; n5 lt	1230
	2; n1 lt; twins dead	10467
Cercopithecus ascanius	1	4182
	1 ea; n2 lt	10505
Cercopithecus cephus	mode 1	6742
Cercopithecus lhoesti	1 emb; n1 lt	2953, 2955
	1 emb; n1 lt	8856
Cercopithecus neglectus	1 yg; 0.4% twins; n272	1277a
Cercopithecus mitis	1 emb; n1 lt	8856
	1 emb; n1 lt	185
	1 emb ea; n2 lt	331
Cercopithecus mona	1 yg	4354a
	1 emb ea; n2 lt	2953, 2955
Cercopithecus nictitans	1 emb; n1 lt	2953
	1 emb or yg ea; n3 f	185
Colobus abyssinicus?	1 emb ea; n2 f	4926
Colobus angolensis	1 ea; n3 lt	2109
Colobus badius	1 emb ea; n2 lt	184
Colobus polykomos	1 yg	6747
	1 emb; n1 lt	8856
Colobus satanas	1 emb; n1 lt	8508
Erythrocebus patas	est 1	5522
	1.01 yg; r1-2; n501 lt	10072
Macaca arctoides	1 yg; n1 lt	11985
	1 yg ea; n26 lt	3107
Macaca fascicularis	1 emb; n1 lt	2313
	1 yg ea; n5 lt	5649
	1 ea; n65 lt	10844
Macaca fuscata	2 yg; n1 lt	7758
Macaca mulatta	1	4546
	1 yg ea; n17 lt	1230
	1 ea; n19 lt	6684
	1 ea; n132 lt, 45 f	264
	1 yg; 0.1% twins; r1-2; n1003 births	5920a
	1.01 yg; r1-2; n580 lt	11199
Macaca nigra	1 yg	4462
Macaca radiata	1	9982
	1 yg ea; n12 lt	1230
Macaca silenus	1; n1 lt	2109
Macaca sinica	mode 1; r1-2	8496, 8501

Macaca sylvanus	1.003 yg; r1-2; n572 lt	8265
	2 yg; n1 lt	1476
	2 yg; n1 lt	9556
Macaca thibetana	1 yg	12093b
Nasalis larvatus	mode 1; 2 occasional	11852
	1.5 emb; r1-2; n2 lt	9667
Papio hamadryas	1 emb ea; n5 lt	12118
	1 yg ea; n7 lt	1230
	1-2	1445
	2 rare	7494
Papio sphinx	1.20 yg; n5 lt	6508
Presbytis aygula	1	557
Presbytis cristata	mode 1	7161
	1 yg ea; n4 f	8215
Presbytis entellus	1	5273
	1	9175
	1 yg	8720
	1 yg ea; n19	1230
	1-2 yg; twins frequent	8020b
Presbytis melalophos	1	7161
Presbytis obscura	1 yg ea; n2 lt, 2 f	482
Presbytis rubicunda	1 yg	2313
Presbytis senex	mode 1; r1-2	8496, 8501
Pygathrix nemaeus	1 yg	4721
	1 yg ea; n4+ lt	1299
SEXUAL MATURITY		
Cercocebus albigena	f: 1st birth: median ~4 yr; from graph	9341
Cercocebus torquatus	f: 1st sex skin: median 32 mo; r30-39; n6	4183
	f: 1st swelling: 36.4 mo; se1.1; n14	4159a
	f: 1st menses: 37.6 mo; se1.3; n12	4159a
	f: 1st conceive: 51 mo; se1.8; n28	4159a
	f: 1st birth: median ~4.5 yr; from graph	9341
	f: 1st birth: 56.5 mo; se1.8; r40.8-81.6; n28	4159a
Cercopithecus aethiops	f: 1st conceive: 4.4-5.7 yr; n3 habitats	6350
	f: 1st birth: 3-4 yr	3152
	f: 1st birth: median ~4 yr; from graph	9341
	f: sexual maturity to 1st preg: 15 mo	401
	f: 34 mo	4995
	m: 60 mo	4995
	f/m: 1st repro: ~40 mo	1225
Cercopithecus mitis	f: 1st birth: median ~5.5 yr; from graph	9341
	f/m: 1st repro: 55 mo; n1	1225
Cercopithecus neglectus	f: 1st birth: mode 4-5 yr	1277a
	f: 1st birth: median 5 yr; from graph	9341
	m: 6-8 yr	1127a
Cercopithecus talapoin	f: menarche: 2.5-3.5 yr	9338
	f: menarche: 36.8 mo; r31-54; n6	9337
	f: 1st conceive: 51.5 mo; r42-58; n4	9337
	f: 1st conceive: 4.5 yr; n3	9338
	m: testes enlarge: 5 yr; n3	9337
Colobus guereza	f: 1st birth: median ~5.75 yr; from graph	9341
Colobus polykomos	f/m: ~2 yr	5758
Erythrocebus patas	f: menarche: 25.7 mo; r19-30.5; n5	9337
	f: 1st conceive: 157 wk; r134-184	10072
	f: 1st conceive: 29 mo; r19-40; n8	1797
	f: 1st birth: 35 mo; r25-46; n8	1797, 9341
	f: 3.5 yr	5758
	m: 1st sire: 3.67 yr	10072
	m: scrotum turns blue: 4 yr; n2	9337
	m: 5 yr	5758
Macaca arctoides	f: progesterone rise: 3.73 yr; r3.42-4.23; 6.2 kg; r5.7-7.0; n6	7908, 7909
	f: 1st conceive: 3.00 yr; n1	10913
	f: 1st conceive: 4.4 yr; n5	3107
	f: 1st birth: 4.90 yr; r4.02-6.55; n8	7909

	m: testes descend: 3.33 yr; r2.76-3.69; 5.7 kg; r5.2-6.3; n9	7908
	m: 1st sire: 3.25 yr; n1	10913
	f/m: est 4-5 yr	3386
Macaca cyclopis	f: menarche: 36.3 mo; sd6.0; r31.4-46.4; n7	8347
Macaca fascicularis	f: menarche: 2.79 yr; n12	10264
	f: 1st mate: 2.25-3.17 yr	4384
	f: conceive: 3.25-5.33 yr	4384
	f: 1st birth: 3.83 yr; r1.0-4.5	7954
	m: spermatogenesis: 44-53 mo	3434
	m: spermatogenesis: 3.9 yr; n6	10264
	m: sperm in ejaculate: 3.5-4.2 yr	4384
	m: testes: large 36-52 mo; 3-4 kg	10359
	m: mate: 3.83-4.5 yr	4384
Macaca fuscata	f: ovarian activity: 2 yr	5156
	f: menarche: 3-3.5 yr	7915
	f: 1st estrus: 3.5-6.5 yr	10633
	f: 1st conceive: 3.5 yr	4315
	f: 1st birth: 3.92 yr	7916
	f: 1st birth: 4.5 yr; r4-6.92	7954
	f: 1st birth: 5.15 yr; sd0.86; r4-7; n27	3187
	f: 1st birth: 5.5-6.4 yr	3980
	m: 1st mount: 2.5 yr	4315
	m: 1st spermatogenesis: 4 yr	7917
	m: 1st ejaculate: 4 yr	9304
	m: 1st ejaculate: 4.5 yr	4315
Macaca mulatta	f: 1st sex skin: 1.851 yr; se0.146; r1.159-2.178; 3316 g; se78; r3100-3500; n6	11693
	f: 1st sex skin: 2-4 yr	9333
	f: 1st sex skin: median 26.5 mo; r24-35; n14	4183
	f: menarche: 1.941 yr; se0.128; r1.480-2.440; 3354 g; se106; r2821-3600; n7	11693
	f: menarche: 722 d; r14-31 mo; ~3 kg	11198, 11199
	f: menarche: 27 mo; 3.7 kg	3427
	f: menarche: 127.3 wk; ±6.6; n22	9056
	f: menarche: 30.3 mo; se3.1; n22	10745
	f: menarche: 133 wk; ±3.0; 3.46 kg; ±0.08; n26	9056
	f: menarche: 30.7 mo; ±1.2; n30	10744
	f: menarche: 30.9 mo; ±1.1; n31	11771
	f: menarche: 31.1 mo; se2.6; n14	10742
	f: menarche: ~3 yr; 3.35 kg	4448, 4546
	f: menarche: 2.9-4.7 kg	173
	f: sexual maturity to 1st preg: 15 mo	401
	f: 1st ovulate: 3.5 yr	8143
	f: 1st ovulate: 31.5 mo; ±0.4; n31	11771
	f: 1st ovulate: 42.9 mo; ±0.2; n15 1st generation	11771
	f: 1st ovulate: 44 mo; ±2	8929
	f: 1st ovulate: 47 mo; se2.6; n14	10742
	f: 1st ovulate: 48.1 mo; ±2.2; n30	10744
	f: 1st ovulate: 51.2 mo; se3.3; n22	10745
	f: 1st conceive: 3.11, 3.13 yr; n2	6800
	f: 1st conceive: 3.5 yr, 75%; 2.5 yr, 14%	11216
	f: 1st conceive: ~5 yr; 5000 g	4448
	f: 1st birth: 32.9-59.4 mo; fall born	11768
	f: 1st birth: 3 yr 156 d-12 yr 74 d	264
	f: 1st birth: 36.6-59.4 mo; spring born	11768
	f: 1st birth: 4.0 yr; ±0.1; mode 4 59%; r3-7; n56	2721
	f: 1st birth: 4 yr	5312a
	f: 4 yr	5919
	m: testes descend: 3.32 yr; n25	11496
	m: testes large: 4.02 yr; n21	11496
	m: sperm: 2.83-3.42 yr	11200
	m: fully active: 4 yr	11200
	m: 1st sire: 3.30 yr; n2	6800
Macaca nemestrina	f: 1st sex skin: median 35 mo; r30-46; n16	4183
Macaca nigra	f: 1st sex skin: median 48.5 mo; r47-50; n4	4183
Macaca radiata	f: 1st birth: 4 yr	9973

	f: socially mature: ~4 yr	9982
	m: socially mature: ~6 yr	9982
	f/m: 1st mate: 2.5-3.5 yr	9982
Macaca silenus	f: 1st mate: 2.83 yr	6464
	f: 1st birth: 3-4.5 yr	6464
	f: 1st birth: 4.5 yr; r3-7.5	5650
	f: 1st birth: 5 yr	4044
	f: 1st birth: 8 yr	5650
	m: 1st mate: 5.5-6 yr	6464
Macaca sinica	f: puberty: 3 yr	8501
	f: 4-5 yr	8496
Macaca sylvanus	f: 1st sex skin: 3.25-3.5 yr; n11f	7199
	f: 1st conceive: 45.3 mo; r30-78; n8	9171
	f: 1st birth: 4.77 yr; ±0.66; r4-7; n62	8266
	f: 1st birth: 4.99 yr; sd1.13; r2.9-7.9	1477
	m: 1st mount: 4-5 yr	7199
Papio hamadryas	f: 1st estrus: 3.5-4 yr	2501
	f: 1st estrus: 4-4.75 yr	9699
	f: 1st estrus: 4-5.5 yr	10507
	f: menarche: ~3.5 yr	3783
	f: menarche: 183 wk; sd11.7; r168-202; n11	3808
	f: menarche: 189 wk; sd20.5; r141-214; n11	3808
	f: menarche: 200 wk	3809
	f: menarche: 4.3 yr; r4-5; n13 f	9970
	f: menarche: 4.5 yr	213
	f: menarche: 4-5.5 yr	212
	f: sex maturity to 1st preg: 10 mo	401
	f: 1st conceive: ~5.5 yr	213
	f: 1st birth: 4.42 yr; r3.75-4.92	7954
	f: 1st birth: 6.1 yr; r5.5-7; n8	9970
	f: 34.8 mo; r21-45; 7.6 kg	401
	f: 46.8 mo	7494
	m: testes descend: 4.8-6.8 yr	9970
	m: testes: large 3-4 yr	1658
	m: 1st mate: 5 yr	2501
	m: acquired 1st f: 8.5-11 yr	9970
	m: 51.6 mo	7494
Papio leucophaeus	f: 1st estrus: 3.5 yr	1042
Presbytis spp	f: menarche: 4 yr; n1	4784
Presbytis entellus	f: 1st estrus: 3.5-4 yr	5272, 5273
	f: menarche: 28.8 mo; r25.2-31.5; n3	3503
	f: menarche: 3 yr 4 mo 29 d; n1	8884
	f: 1st conceive: 34.2 mo; r31.7-36.9; n3	3503
	m: spermatogenesis; 4.5 yr	8884
	m: 1st mount: 4-6 yr	5272, 5273
Presbytis johnii/P senex	m: min 4 yr	4783
Presbytis senex	f/m: ~3 yr	4781
Theropithecus gelada	f: 1st birth: 4-4.5 yr	2796, 2797
	f: 5 yr	6970
	m: puberty: 3-4 yr, sexually mature: 8-9 yr	2797
CYCLE LENGTH		
Cercocebus albigena	follicular phase: 16.5 d; se4; n4	1553a
	luteal phase: 14.8 d; se1; n4	1553a
	3.4 wk; r2.5-5; n4 cycles, 1 f; from graph	12117
	26.0 d; se0.8; r21-40; n26 intermenstrual intervals, 3 f	1553a
	median 29 d; mode 30; r21-36; n47 cycles, 3 f	9339
	29.75 d; r25-42; n44 cycles, 4 f	3643
	31.3 d; se5; n4 cycles; between LH peaks	1553a
	38.5 d; sd63; r29-?; n8 cycles; 45 d; sd19.6; r29-98; n9 cycles	11428
	47 d; r8-184; n26 cycles; between peak swellings	11428
Cercocebus galeritus	29.83 d; r22-39; n6 cycles, 2 f	3643

Cercocebus torquatus	30 d; r26-33; n5 cycles, 1 f	12117
	30.3 d; r30-31; n3 cycles, 1 f	12121
	32.5 d; r26-35; n8 cycles, 2 f (7 from 1 f)	3643
	33.0 d; n40 cycles, 1 f	7851
	33.4 d; se1.1; mode 30; r28-46; n22 cycles	12123
	median 34.5 d; r25-47 of medians; r10-177; n145 cycles, 14 f	4183
	45.5 d; r26-76	10291
Cercopithecus aethiops	follicular phase: 15.4 d; ±2.4	6081
	estrus: 0-66 d	10503
	luteal phase: 27.8 d; ±4.6	6081
	menses: 3.7 d	11062
	menses: 4.4 d; ±2.1; r1-12; n243	4699
	menses: 4.8 d	2989
	menses: 4.9 d; se0.2; r3-6; n24 cycles, ≤17 f	2967a
	27 d; r24-30	6750
	30.9 d; sd9.6; r16-50	5364
	30.9 d; n392 cycles; cycles >50 d excluded	11062
	31.4; sd5.3; r18-49; n183 cycles, 21 f	4699
	32 d; r18-50; n524 cycles, 28 f	2989
	32.4 d; se0.9; r29-45; n24 cycles, ≤17 f	2967a
	32.5 d; most r18-50; n692	2989
	median 33 d; r25-46; n32 cycles, 5 f	9334
	43.2 d; n16 f	6081
	72.3 d; ±36.23; r24-149; n29 cycles, 7 f	1492
Cercopithecus mitis	menses: 5.3 d; sd1.9; r1-10; n59 cycles, 5 f	2988
	median 30 d; r19-57; n72 cycles, 6 f	9334
	31.9 d; sd6.0; r24-48; n59 cycles, 5 f	2988
Cercopithecus nictitans	28 d; r27-54; n8 cycles, 1 f	3643
Cercopithecus talapoin	luteal phase: 14 d	9338
	menses: 1-4 d	10865
	mode 31 d; median 35; r27-83; n43 cycles, 10 f	9338
	32.9 d; se6.2; r28.0-37.7; n64 cycles, 14 f	9701
	33.0 d; se1.6; r29.1-36.8; n7 f	9701
	33.2 d; r27-43; n5 cycles	10865
	36 d; r29-53; n12 cycles, 2 f	3643
Colobus polykomos	estrus: 1-3 d	3532
Erythrocebus patas	estrus: 12 d; r1-66	6617
	estrus: 13.5 d; ±4.1; n80	6618
	menses: 3 d; r1-5; n16	10072
	30.6 d; sd2.6; r24-38; n64 cycles	10072
	31.9; r10-78	6617
	32.9 d; se1; n82 cycles	6618
Macaca arctoides	follicular phase: 11.9 d; se0.5; r10-14	11707
	luteal phase: 17.2 d; se0.4; r15-19	11707
	menses: 2.8 d	1385
	menses: 3.9 d; ±0.2; n167	6687
	r21-49 d; n10 cycles, 2 f	10912
	mode 28-34 d; r21-34; n1097 cycles, 58 f	10371
	29-30.5 d; across mo	2783
	29.1 d; se0.6; r27-32; n9 cycles, 5 f	11707
	29.4 d; n30 cycles, 6 f	10914
	29.9 d; sd4.4; r19-49; n1365 cycles, 95 f	1385, 2785
	30.2 d; mode 28; r21-42; n189 cycles	7909
	30.7 d; se0.4; n132 cycles, 11 f	6687
Macaca assamensis	follicular phase: 16 d; se1; r9-22; n14 f	11538
	luteal phase: 16 d; se1; most 15-19; n14 f	11538
	r25-35 d; n10 f	11537
	32 d; se2; n14 f	11538
Macaca cyclopis	menses: 4.2 d; sd3.2; mode 3; r1-16; n44 menses	8347
	29.4 d; sd15.5; mode 24; r6-109; n398 cycles, 54 f	8347
Macaca fascicularis	follicular phase: 14.1 d; sd3.11	9863
	follicular phase: 14.4 d; se0.8	2783, 8976, 8977
	luteal phase: 14.2 d; ±0.4; n8 f	7172
	luteal phase: 16.1 d; se0.50	2783, 8976, 8977
	luteal phase: 17.3 d; sd5.8	9863
	menses: 1-6 d	9863

	menses: ~2 d	9446
	menses: 2.8 d; sd1.6; mode 2; r1-9; n707 menses, 28 f	2781, 2782
	menses: 3-5 d 85%; r1-8 d	9827
	menses: 3-6 d	4384
	menses: 3.9 d; mode 4; r1-11; n509 menses, 87 f	11064
	menses: 4 d; ±1	11549
	menses: 4.2 d; se0.3; n35	6687
	menses: mode 6 d; r2-13	2056
	mode 27-28 d; r13-180; n272 cycles	3550
	25-45 d	5316
	28-30 d	11164
	28 d; mode 28-30 70%; r26-38; n20 f	7782
	28 d; median 28; r23-42	4384
	29-32 d	2783
	29.1 d; r26-31; n6 cycles, 2 f	10214
	29.2 d; r26-30; n18 cycles, 15 f	7511
	29.2 d; r22-37; n647 cycles, 93 f	9827
	29.4 d; sd4.3; mode 27, 38; r20-42; n240 cycles, 60 f; 28.8 d; sd2.9; without long cycles	12027
	30.3 d; sd4.1	1280
	30.9 d; sd4.8; median 33; mode 28; r19-43; n595 cycles, 28 f	2781, 2782, 2785, 2786, 5310, 8976, 8977
	31.0 d; sd1.7; n8 f	7172
	31 d; sd1; n25 f	11549
	31 d; median 31; mode 27; r23-41; n345 cycles, 19 f	10264
	31.3 d; se1.5; n21 cycles, 11 f	6687
	31.4 d; n17 cycles, (15/17 ovulatory)	9863
	31.5 d; se1.3; median 32, 36; n77 cycles, 30 f	9446
	32.5 d; se2.3; n6 cycles	7473
	mode 33 d; most 25-42; r15-50; n372 cycles	11062, 11064
	34.5 d; se1.3; r27-82; n275 cycles, 55 f	2247
	39.2 d; mode 32-34; r24-111; n62 cycles, 4 f	2056
	41 d; n4 cycles, 1 f	12117
	47.2 d; mode 26-28; r26-30 26%; median 32; r9-339; n326 cycles, 45 f	5687
Macaca fuscata	follicular phase: 12.2 d; ±1.4; r10-15; n21 cycles, 20 f	7918
	estrus: 8.0-8.9 d	11873
	luteal phase: 8.2 d; ±1.6; r6-10; n5 no progesterone rise	7918
	luteal phase: 14.0 d; ±1.1; r13-16; n13 progesterone rise	7918
	menses: 2-4 d	7954
	menses: 3.6 d	8139
	24-28 d	7954
	25 d; n2 cycles, 2 f	4516
	26.3 d; sd5.4	7915
	4 wk	429
	28.5 d; sd3.0; r23-36; n36 cycles, 7 f	8139
	29.39 d; sd16.9; n129 cycles, 70 f	11873
	30 d; n1 f	4315
	34.1 d; ±24.8; mode 26 d; r11-107; 30.6 d; ±22.8; excluding >60 d	10633
Macaca maura	20-40 d	8885
	median 32 d; r27-37 of medians; r14-85; n33 cycles, 5 f	4183
Macaca mulatta	follicular phase: 10.6 d; ±1.4	1131
	follicular phase: 12 d	12084
	follicular phase: 12.59 d; ±3.13	6732
	follicular phase: 14.1 d; ±0.6	455
	estrus: 9.4 d; ±5.1; r1-25	6466, 6467
	luteal phase: 14.85 d; ±1.23	6732
	luteal phase: 14.9 d; ±2.1	1131
	luteal phase: 15 d	12084
	luteal phase: 15.8 d; ±0.3	455

	menses: 2.7 d; ±1.0; n30 f	618
	menses: 2.8 d; ±0.2; n8 cycles	914
	menses: 3.3 d; ±0.1; n82	6687
	menses: 3.4 d	4534
	menses: 3.6 d; r1.7-6.6; n520 cycles, 30 f	4448
	menses: 3.9 d; mode 4; r1-11; r1-8 with conception	11064
	menses: mode 4 d; r2-8	173
	menses: mode 5 d; r2-11; n132, 10 f	2055
	20.4 d; sd9.2; r6-45; n204 cycles, 20 f	6616
	r24-30 d; n39 cycles, 13 f	7819
	r24.6-33.0 d per f; est r14-42; n481 cycles, 30 f	4448
	r25-27 d; n36 cycles, 12 f	7816
	r25-31 d; n23 f	8807, 10265
	r25-35 d; n9 f	11537
	r25-35 d; n25 f	1696
	mode 25-28 d; r10-70; n15 cycles	173
	25.5 d; sd2.3; r22-30; n8 cycles, 8 f	1131
	r26-30 d; n10 cycles, 10 f	12084
	26.27 d; r23-33; n35 cycles, 7 f	661
	26.5 d; n47 cycles, 15 f	4829
	26.7 d; sd2.1; r24-31; n12 cycles, 9 f	11423
	26.9 d; sd4.9; mode 27-28; r24-30 78%; r19-40; n18 cycles	1132
	mode 27 d; most 15-51; r15-190; n125 cycles, 10 f	2055
	27-28 d; mode 28; r14-42; n392 cycles, 22 f	4448
	27.25 d; se~1.5; n15 cycles, 5 f	2977
	27.36 d; r21-36; mode 27; sd3.18; n33 cycles, 13 ? f	6732
	27.5 d; se0.68; n19 cycles, 4 f	7818
	27.6 d; se1.0; n14 cycles, 7 f	7472
	mode 28 d; r21-37	8649
	mode 28, 30 d; most 23-35; r15-50; n2047 cycles	11062, 11064
	28.3 d; r23-41; n12 cycles, 1 f	12117
	28.4 d; sd2.56; mode 28; n101 cycles, 7 f	3065
	28.5 d; se2.18; r25-34; n2	4534
	28.5 d; se2.2; r22-28; n30 f	618
	29 d; sd3; n450 cycles, 30 f	8509
	29.2 d; se0.7; n74 cycles, 27 f	6687
	29.4 d; sd6.6; median 24; n30 cycles	6467
	29.5 d; sd5.01; r24-50; n39 cycles, 31 f	2195
	29.9 d; se0.5; n9 f	455
	median 30; spring; median 35; winter	9332
	30.6 d; r23-53; n44 cycles, 15 f	3064
	30.6 d; n183 cycles, 16 f; 35.4 d anovulatory cycles	11421
	30.7 d; n24 summer cycles; 26.2 d; n30 winter cycles, 9 f	9132
	30.8 d; mode 28; se0.4; r6-237; n1256 cycles, 50 f	2219
	31.5 d; mode 28; se1.18; n167 cycles	12123
	35.4 d; median 30d; mode 28; 35% r26-30; r14-192; n523 cycles, 65 f	5687
Macaca nemestrina	follicular phase: 13.9 d; r11-60; n23; perineal detumescence	11641
	follicular phase: 14.8 d; r11.5-21.5; n10; laparoscopy	11641
	follicular phase: 15.1 d; ±3.1	975
	follicular phase: 15.5 d; ±0.7; 18.6 d; ±4.5; 2 measures	2870
	follicular phase: 16.2 d; se0.47; r11-35; n25	12123
	follicular phase: 17.6-19.2 d; n21 cycles	986
	luteal phase: 13.6-15.2 d; n21 cycles	986
	luteal phase: 14.3 d; ±1.1	975
	luteal phase: 14.4 d; r4-19; n23; perineal detumescence	11641
	luteal phase: 14.6 d; r13-16.5; n10; laparoscopy	11641
	luteal phase: 15.1 d; se0.31; r11-18; n25	12123
	menses: 5 d; mode 5; r3-7; n22	11641
	29.2 d; n9 cycles, 5 f	975
	31.8 d; mode 28-29; r17-75; n28 cycles, 5 f	11641
	32.08 d; mode 30; sd0.8; r24-66	6177
	32.5 d; se1.5; mode 29; r27-68; n30 cycles, 5 f	12123
	32.8 d; n21 cycles	986
	34.2 d; sd4.5; n47 cycles, 5 f	2870

	median 35.25 d; r of medians 14-77; r6-396; n485 cycles, 32 f	4183
	38.2 d; r31-47; n5 cycles, 1 f	12117
	40.1 d; ±13.4; mode 31; r11-96; n69 cycles, 24 f	6083
Macaca nigra	menses: 4.6 d; n5	2620
	30.7 d; r28-35; n6 cycles, 1 f	10264
	~31 d	1016
	33.5 d; r23-45; n2 cycles, 3 f	2620
	median 35.5 d; r30-44.5 of medians; r10-115; n159 cycles, 12 f	4183
	median 40 d; n15 cycles, 7 f	838
Macaca radiata	menses: 3.8 d; mode 3; n101 menses	11064
	r25-27 d; n6 cycles, 4 f	7511
	r25-36 d; n4 cycles	4450
	26-27 d	10290
	27.06 d; r23-31; n15 cycles, 6 f	6228
	28 d	7009
	mode 28, 30 d; n76 cycles	11062, 11064
	28.0 d; se0.56; n6 f	8971
	29.5 d; mode 31; r25-38	6177
	29.6 d; r10-57; n62 cycles, 22 f	607
	33 d; 44 d with 2 long cycles	12117
Macaca silenus	follicular phase: 14 d	8889
	estrus: 9 d	2497
	luteal phase: 16 d	8889
	28 d	2497
	30 d; r30-36; n2 zoos	8889
	30 d; n8 cycles, 5 f	9908
	31 d; se0.63; n10 cycles, 4 f	9904
Macaca sinica	29.5 d; mode 31; r25-38	6177
Macaca sylvanus	menses: 3-4 d	8223
	27-33 d	8223
Papio hamadryas	follicular phase: 15.7 d; ±0.52; r6-25	9826
	follicular phase: 17.1 d; se0.35; n72	12123, 12129
	follicular phase: 21.8 d; se1.49; n55	12123
	follicular phase: 24.5 d	9699
	luteal phase: 14.4 d; se0.25; n55	12123
	luteal phase: 15 d	9699
	luteal phase: 15.1 d; se0.24; n72	12123, 12129
	luteal phase: 15.6 d; ±0.35; r10-20	9826
	luteal phase: 15.8 d; ±0.4	11599
	luteal phase: 18 d	128
	menses: 1.4 d	10147
	menses: 3 d	12129
	menses: 3 d; r1-6	4629
	menses: 3.7 d	10836
	menses: mode 4 d; median 4; r1-7	913
	13.78 d; r3-84; n29 perineal cycles, 19 f	7331
	23-37 d	916
	median 35 d; mode 36; r29-39; n32 cycles, 8 f	913
	29.3 d	690
	29 d; n10 f	5989
	30 d; n1 cycle, 1 f	12118
	30-33 d; n17 f	3884
	30.3 d; n100 cycles, 20 f	10147
	30.3 d; se0.22; n200 cycles	10385
	30.6 d; r23-40; n3 cycles, 8 f	12117
	30.9 d; se0.95; r25-37; n11 cycles, 1 f	12127
	31.4 d; se1.0; sd5.3; r23-47; n26 cycles, 5 f	4507
	31.4 d; se0.41; n72 cycles, 10 f	12129
	31.5 d; r20-38; n53 cycles, 53 f	9826
	32 d; r24-38; n300 f	3885
	32.98 d; se~0.9; n30 cycles	10836
	33.16 d; sd3.74; mode 31; r19-43; n96 cycles, 32 f	4628, 4629
	33.3 d; se0.7; mode 31-32; r25-41; n32 cycles, 2 f	12123
	34.5 d; n41 cycles, 11 f	11209

	34.6 d; r28-46; n5 f	3651
	34.7 d; se0.48; r30-41; n25 cycles, 1 f	12117, 12123, 12127
	35.61 d; sd3.206; se0.158; mode 35; r29-42; n404 cycles, 34 f	3783
	35.7 d; se0.66; n40 cycles, 11 f	11682
	36 d; r18-52; n150 cycles, 40 f	4627
	36.2 d; se1.5; mode 33; n55 cycles, 5 f	12117, 12123
	38.8 d; r38-42; n~144 cycles, 16 f	3782
	40.11 d; r33-65; n9 cycles, 1 f	2591
	40 d	9699
Papio leucophaeus	r25-42 d; n12 cycles, 3 f	1042
	30.8 d; n12 cycles, 1 f	7851
	32.6 d; se0.9; n18 cycles, 2 f	12123
	median 35 d; r30-46.5 of medians; r13-82; n88 cycles, 3 f	4183
Papio sphinx	32.0 d; n55 cycles, 3 f	7851
	median 33.5 d; r32-35 of medians; r20-244; n46 cycles, 2 f	4183
Presbytis entellus	estrus: 5-7 d	5272, 8119
	menses: 2.0 d; ±0.1; r1-3	6547
	menses: 2-4 d	2293
	menses: mode 2-4 d; r1-11; n86, 16 f	2292, 9833
	21 d; n4 cycles, 2 f	9833
	24.1 d; sd3.8; r17-36; n37 cycles, 9 f	3503
	26.3 d; sd3.6; median 22; r18-45; n125 cycles, 5 f	6547
	26.8 d; sd9.1; mode 22; r21-26 68%; r19-68; n76 cycles, 16 f	2292
	28 d; r24-34; n10 cycles	5044
	30 d	8119
Presbytis obscura	estrus: 3 d	6369
Pygathrix nemaeus	estrus: 2-3 d	4720
	3 wk	4720
	28-30 d	7983
Theropithecus gelada	estrus: 7-10 d	2799
	estrus: 8-10 d	1015
	menses: 3 d	1015
	29 d; r24-34; n2 f	2799
	34.6 d; sd2.3; r31-39; n22 cycles, 1 f?	10142, 10143
	~35 d	2796
	35.4 d	1015
GESTATION		
Cercocebus albigena	176 d; r174-179; last swelling to birth	9339
	177.3 d; r170-181; n3	3643
	184-189 d; n3	11428
Cercocebus galeritus	180 d; se4.49	5768a
Cercocebus torquatus	160-174 d	10291
	161.3 d; r160-162; n3	3643
	median 167 d; r155-199; n23 birth, 12 f	4183
	170.7 d; r164-175; n3 births, 2 f; sex skin swelling to birth	10387
Cercopithecus aethiops	implant: 15-21 d postcoitus	11062
	140-160 d	5758
	153 d	9341
	163 d; ±2	1225
	163.2 d; sd6.3; n38	5364
	165 d; r142-171; n11	9334
	175-178 d	6750
	~7.5 mo	7485
Cercopithecus ascanius	est 120-130 d	5758
	148.5 d; r122-175; n2	2047
	5-6 mo	3581
Cercopithecus mitis	120-130 d	5758
	r138-140 d; n4; (127 d; stillborn gestation not used)	9334
Cercopithecus neglectus	est 170 d	9341
	178 d; r126-229	1127a
Cercopithecus talapoin	156 d	9335
	158-166 d; n18	9338
	158 d; ±5; live birth	9701
	164 d; ±5; n2 stillborn	9701
	167 d; n1	3643

	est ~7 mo; field data	3640
Colobus badius	est 4.5-5.5 mo; field data	10503a
Colobus guereza	est 170 d	9341
Colobus polykomos	174.7 d; r169-178; n3	1956
	est 195-225 d	3532
Erythrocebus patas	167 d	9341
	167 d; se2.2; r159-174; n8	6618
	167.2 d; sd10.3; r133-190; n142	10072
	est 170 d; n1	3966
Macaca arctoides	implant: 16.8 d; se1.8; n5	6687
	172.4 d; n8	1385
	174.7 d; sd8.3; n9	2785
	175 d; r161-186; n12	10371
	176.6 d; r167-185; n25	7909
	177.5 d; se2.1; n10 lt	6687
	178 d; n5 timed	10912, 10914
	207 d; n1	11985
Macaca cyclopis	162 d; sd9.6; r142-175; n9	8347
Macaca fascicularis	implant: 17 d; ±4	11549
	implant: 19 d; r18-21; n3	5310
	implant: 19.2 d; se0.8; n14	6687
	160 d; sd3; n8	11549
	160 d; sd7; n109	11062, 11064
	160-162 or 224 d; est 160-162 is correct	10263
	mode 161-170 d; r151-196	3550
	162.7 d; se0.9; n10	6687
	164 d; r139-176; n101	4954
	165.4 d; sd2.2; n5	2785, 2786, 5310
	167 d; r153-179; n17 after last mating	10264
	169 d	8977
	180 d; r167-193; n31 after last menses	10264
Macaca fuscata	~5 mo; field data	5974
	150-165 d	7405
	171-180 d	5589
	173 d; sd6.9; r161-188; n17	7916
	~7 mo; field data	5139
Macaca mulatta	implant: 9 d; n1	9023
	implant: 9-10 d	11861
	implant: 9-11 d	4707
	implant: 9.5 d	3017
	implant: 10.5 d	1109
	implant: 20.2 d; se0.9; n13	6687
	implant: 21.4 d; ±7.29	6467
	135-171 d	3545
	139 d	6880
	~5 mo	10234
	150 d; n24	5228
	r152-173 d; n18; timed	8588
	r155-172 d; n10	4854
	r157-168 d; n6	7380
	r158-173 d; n16	6038
	159-174 d; n3	4445
	161.2 d; r145-169; n6	49
	164 d; ±4; r153-176	6880
	164 d; n38	4448
	164 d; sd6; n967	11061, 11062, 11064
	165 d; r162-169; n4	2557
	165.7 d viable birth	4448
	165.7 d; se1.5; n12	6687
	166.6 d	914
	167 d; ±7	8509
	167 d; ±6; r142-181; n441 wild	6880
	168.2 d; n5 primiparous	4448
	168.4 d; se1.3; n8	8589
	168.9 d; sd6.85; r148-194; n225	11201

	168.98 d; n57	11408
	169 d; 90% 156-179; r140-195; n268	5104, 11199
	170-175 d	1680
	~177 d	5312a
Macaca nemestrina	170 d; mode 168; r162-186; n21	6083
	median 171 d; r154-230; n56 birth, 23 f	4183
	171 d; n1	12119
	172.5 d; sd6.4; r154-189; n90	9373
Macaca nigra	170.9 d; mode 173; r165-190; n16; 142, 206 d excluded	2153
	median 176 d; r166-181; n15 birth, 8 f	4183
	≤ 7 mo	4462
Macaca radiata	153-169 d; n3	4450
	162 d; sd4; n16	11062, 11064
	169 d; r166.5-170.5; n4	10290
Macaca silenus	171.8 d; r166-176; n4	6464
	172 d; r154-186	8889
	173-188 d; n2	2495, 2497
Macaca sinica	~7 wk	8496
	~6 mo	8501
Macaca sylvanus	implant: 18.2 d	6127
	164.7 d; sd6.11; median 165; r145-177; n33	6127
	5-6 mo	6711
	163.2 d; sd5.38; r145-177; n56; m offspring	8265
	165.3 d; sd4.55; r145-178; n67; f offspring	8265
Papio spp	175 d; ±11; r164-186	5012, 6024
Papio hamadryas	implant: 9 d	5011
	154-185 d; sex skin detumescence to birth	12118
	5.5-7 mo	4582
	r164-179 d; n4	8357
	175 d; r164-186	7494
	25 wk; n23; sex skin detumescence to birth	213
	169-185 d; n3	7233
	175-185 d	4870
	180 d; se0.98; n13	10162a
	181 d; se4; n13	8358
	184 d; n1	3783
	187 d; r173-193; n14 lt, 11 f	3776
Papio leucophaeus	173 d; n1	4183
	181.3 d; r179-183; n3	1042
Papio sphinx	4.5 mo	4582
	median 175 d; r164-213; n4 birth, 1 f	4183
	183-225 d; n4 births, 2 f; sex skin swelling to birth	6508
Presbytis entellus	165-185 d	8119
	~180-196 d	9175
	6-7 mo	4377
	200.1 d; r196-202; n7; field data	11819
Presbytis obscura	140-150 d	6369
	5 mo	482
Presbytis senex	~6 mo	8501
	6.5-8.5 mo	9361
Pygathrix nemaeus	~165 d; n7	4723
	r180-190 d; n8	1299, 6503
Theropithecus gelada	~5.5 mo	2799
	6 mo	2796
	~200 d; n2	10142
LACTATION		
Allenopithecus nigroviridis	solid food: 2.5 mo	8702
	wean: 2.5 mo	8705
Cercocebus albigena	wean: 140 d	9339
	wean: 33.33 mo; ±15.87; r17-48	11428
Cercopithecus aethiops	nurse: max 6 mo	3151
	nurse: 6 mo	9728
	wean: 6 mo	7485
	nurse: 206 d; n4	4995

	nurse: 263 d; ±3	10503
Cercopithecus ascanius	independent: ~6 mo	4182
	solid food: 2 mo	4182
Cercopithecus mitis	solid food: 1.5-2 mo	146
	independent: 2 mo	10158
Cercopithecus talapoin	solid food: ~1 mo	3642
	wean: 49-210 d	9338
Colobus polykomos	solid food: 4-5 wk	3532
	wean: 6 mo	3532
Erythrocebus patas	wean: 6 wk; n1	3966
Macaca arctoides	wean: 6-15 mo	10914
	wean: 7 mo	7909
	wean: 10-14 mo; n4	10912
Macaca fascicularis	nurse: 84 d; ±10.4; r52-127	2248
	wean: 100-120 d	4839
	wean: 207 d; sd71; r96-286; n12	11595
	wean: 245 d; sd30; r184-295; n16	11595
	wean: 9-18 mo	10264
Macaca fuscata	solid food: 3 mo	10667
	wean: est 6 mo	10667
Macaca mulatta	solid food: ~2 wk	6466
	solid food; 2 mo	4448
	wean: 7-14 mo	4448
	wean: 6-9 mo	8722
	wean: 8 mo	6800
	wean: ~10 mo	6466
	wean: ~1 yr	10234
Macaca nigra	solid food: 5 wk	4462
	wean: ~1 yr	4462
Macaca radiata	wean: 8-12 mo	9982
Macaca silenus	solid food: 20 d	2495, 2497
	solid food: 4 wk	2109
	wean: 1 yr	2495, 2497
Macaca thibetana	wean: ~6 mo	3386
Nasalis larvatus	solid food: 6 wk	8710
	wean: 7 mo	8710
Papio spp	wean: 14-18 mo	6342
Papio hamadryas	solid food: 33 d	3776
	wean: 6-8 mo	10158
	wean: ~6-~15 mo	5758
	wean: ~12 mo	1486
Papio leucophaeus	wean: 16 mo	1042
Presbytis entellus	solid food: 2-3 mo	5273
	wean: 8-10 mo	9175
	wean: 12-15 mo	5273
	wean: ~15-16 mo	4377
Presbytis senex	wean: 28 wk	9361
Theropithecus gelada	wean: 12-15 mo	10142
	wean: 12-18 mo	2799
INTERBIRTH INTERVAL		
Cercocebus albigena	median ~1.6 yr; n6; from graph	9341
Cercocebus torquatus	median ~1.3 yr; n9; from graph	9341
	median 390 d; r120-780; n19 intervals, 10 f	4183
	16.6 mo; se0.6; n110	4159a
Cercopithecus spp	3 yr	3748
Cercopithecus aethiops	10.7 mo; r5-24; n57	3151
	337.9 d; n29	1225
	11.7 mo; se0.4; with new m	3153
	353 d; yg survives	4995
	median ~1 yr; n42; from graph	9341
	12.15 mo; r8.75-19; n5	6750
	373.8 d; r223-512; n55 intervals	9728
	13 mo; r10-16; n3	7485
	13.6 mo; se0.8; with old m	3153

Cercopithecus mitis	413 d; n11	1225
	median ~1.5 yr; n30; from graph	9341
	19.2 mo; r14.4-25.8	2988
Cercopithecus neglectus	mode 12 mo	1127a
	median ~1.6 yr; n17; from graph	9341
Colobus badius	8-28 mo; field data	10503a
Colobus guereza	median ~1 yr; n16; from graph	9341
Colobus polykomos	14 mo; r12.5-15.5; n2	3532
	459 d; r283-731; n6 intervals, 4 f	1956
Erythrocebus patas	348.4 d; se16.4	6618
	11.8 mo; n31 intervals, 9 f	1797
	median ~1 yr; n11; from graph	9341
Macaca arctoides	523.1 d; r412-691; n17 intervals, 10 f; indoor	7909
	median 525 d; r390-1080; n10 intervals, 5 f	4183
	544.6 d; se36.5; r363-908; yg lives	4464
	619.4 d; r444-910; n21 intervals, 13 f; outdoor	7909
	19 mo 2 d; n9 f	3107
Macaca fascicularis	median 390 d; r330-720; n22 intervals, 11 f	4183
	448 d; r335-626	11595
	~15 mo; n832	7954
Macaca fuscata	274-448 d; n53; 564-796; n107; >796-1115; n8	10667
	median 386 d; mode 425.3; sd113.5; r325-833; 2 outliers 679, 833 d; n29	9702
	2 yr; n283	5974, 7954
Macaca mulatta	1 yr; n94; 2 yr; n45	10077
	median 360 d; r300-690; n29 intervals, 23 f	4183
	362 d; r294-441; n2	5312a
	406 d; se23; n19 high rank	3883
	15 mo	4546
	17.3 mo	401
	630 d; se94; n11 low rank	3883
Macaca nemestrina	median 405 d; r240-1080; n44 intervals, 13 f	4183
Macaca nigra	median 540 d; r300-720; n13 intervals, 7 f	4183
Macaca sylvanus	210.5 d; sd29.1; median 203; r183-310; n21	6127
	1 yr; 88.4%; 2 yr; 11.6%	8266
	431.716 d; sd165.941; median 373.75; mode 361; ~1 yr 85.2%; r231-1121; n155 ibi	1477
	31.5 mo; r22-49; n8; yg lives	9171
Macaca thibetana	2 yr	12093b
Papio hamadryas	11 mo; stillborn	213
	12-16 mo; forest; 18-24 mo; savanna	5758
	423.5 d; r382-465; n2 samples	6937
	~16.5 mo	401
	18 mo; yg lives	1486
	592-840 d; yg lives	10507
	?-22 mo; yg lives	213
	24 mo; n16 intervals, 12 f; yg lives	9970
	760 d; se34.68; n23	10162a
Papio leucophaeus	19 mo	1042
Presbytis entellus	14.1 mo; r8.2-20.2; n25 ibi, 13 f	3503
	15.3 mo; r11.2-20.2; n19 ibi, 11 f; yg lives	3503
	15.4 mo; yg lives	4377
	15.7 mo; n14 ibi, 9 f; yg lives	4377a
	2 yr	5273
Presbytis senex	16-24 mo	9361
Pygathrix nemaeus	1 yr	6502
Theropithecus gelada	median 525 d; r270-720; n11 intervals, 6 f	4183
	est 2-3 yr	2796
	2.14 yr; r1-6 yr; 1.64 yr if harem taken over	2797

SEASONALITY

Cercocebus albigena	birth: May, July, Sept, Dec; n2 f; zoo	2109
	birth: yr rd; W Africa	5758
Cercocebus galeritus	est Dec-Feb; ne Gabon	8820
Cercocebus torquatus	sex skin: Apr-early June; n1; zoo	12117
	birth: Apr-Feb; n32; captive	4183
	birth: May, July, Aug, Dec; few data; zoo	1230
	birth: yr rd, peak Oct-Mar; captive	4159a
Cercopithecus spp	preg f: July-Oct; Kisangani, Zaire	3748
	birth: Jan, Apr-Nov; n17; captive	537
	birth: July-Oct, peak Aug; Kisangani, Zaire	3748
Cercopithecus aethiops	testes: large June-Sept; captive	2967
	testosterone: high Dec-Mar, low Apr-Nov; captive	2967
	cycles: normal May-Nov; prolonged Jan-June; lab	2989
	mate: mid Mar; zoo	7485
	mate: May-mid Oct; Kenya	10503
	mate: June-July; Kenya	256
	mate: est yr rd; W Indies	2009
	mate: yr rd; e Africa	6612
	conceive: May-June; n Kenya	11654
	conceive: yr rd; captive	5364
	preg f: Feb-Mar; n7; Ethiopia	10311
	preg f: yr rd; Botswana	10154
	birth: Jan-Feb; Buffle Noir, Cameroun	5587
	birth: Jan-June; n11/12; July; n1/12; captive	9334
	birth: Feb, Sept-Nov; few data; zoo	1230
	birth: July-Aug; Kalamaloue, Cameroun	5587
	birth: peak Aug-Oct; Zambia	331
	birth: Sept-May; Zaire	11253
	birth: Oct-Nov; Bakosi, Cameroun	5587
	birth: Oct-Mar; Kenya	10503, 10504
	birth: Nov; zoo	7485
	birth: yr rd, peak Apr-July 73%; W Indies	4995
	birth: yr rd, peak May-Sept; captive	6126
	birth: yr rd; captive	3151
	birth: yr rd; S Africa	10158
Cercopithecus ascanius	estrus: yr rd, peak May-Sept; w Kenya	2047
	mate: Mar-Apr; Central Republic of Africa	3581
	mate: Dec-Apr; Uganda	5758
	mate: peak Dec-Apr; small emb Jan-May; Uganda	4182
	birth: Jan, Sept-Oct, Dec; Kenya	2047
	birth: Mar-Apr, June, Oct-Dec; few data; Uganda	10505
	birth: peaks May, Sept; Uganda	5758
	birth: Aug-Sept; Central Republic of Africa	3581
Cercopithecus campbelli	mate: est June-July; Ivory Coast	1148
	birth: end wet-early dry season; Ivory Coast	1148
Cercopithecus cephus	birth: Dec-Mar; n9; Gabon	3640
Cercopithecus lhoesti	preg f: Feb; few data; Cameroun	2955
	lact f: Feb, Oct, Nov; few data; Cameroun	2955
Cercopithecus mitis	mate: Jan, Mar, July-Dec; Uganda	146
	mate: Aug-Jan; Kenya	8106
	mate: yr rd; Rift Valley, Kenya	1116
	conceive: July-Nov; Kenya	8107
	conceive: peak June-Oct; Kenya	2048
	birth: Jan, May, July-Sept, Nov; n13; Uganda	146
	birth: June-July, Nov; n5; captive	9334
	birth: est Sept-Apr; S Africa	10158
	birth: Sept-Dec; Zimbabwe, Zambia, Malawi	10153
	birth: Nov-Mar; Kenya	8107
	birth: Dec-May; n14; Uganda	9362
	birth: est yr rd; captive	2988
	birth: yr rd; Kenya	2048
Cercopithecus mona	birth: Dec-Apr; n5; Gabon	3640

Cercopithecus neglectus	conceive: May-July, Oct-Nov; s Cameroun, Benito, Equatorial Guinea	647
	birth: Feb-Mar; n3; Gabon	3640
Cercopithecus nictitans	preg f: Mar; n1; e Africa	185
	repro: Mar; n1; Cameroun	2955
Cercopithecus talapoin	mate: Jan-Mar, peak Feb; Cameroun	9340
	mate: May-Aug, peak July; n7; Gabon	3640
	mate: Dec-Feb; Cameroun	9335
	conceive: yr rd; captive	9338
	preg f: Oct-Dec; Cameroun	9340
	birth: June-Aug; Cameroun	9340
	birth: Aug-Sept; Cameroun	9335
	birth: peak Nov-Mar; Gabon	3639
	birth: Nov-Apr; Gabon	3641
	birth: Dec-Apr; n18; Gabon	3640
Colobus abyssinicus?	preg f: June, Oct; n2; Kenya	4926
	birth: yr rd; Uganda	5758
Colobus angolensis	birth: Feb-Dec; few data; zoo	2109
Colobus badius	preg f: Sept; n2; Liberia	184
	birth: yr rd; n27; Kenya	6833
	lact f: Dec; n2; Tanzania	186
Colobus polykomos	mate: yr rd; Zaire	11271
	birth: Sept-Apr; Eq Guinea	8505
	birth: yr rd; e Africa	5758
Erythrocebus patas	mate: May-Dec; Kenya	1796
	mate: June-July; Uganda	4232
	conceive: June-Aug; Kenya	1796
	conceive: June-Aug; Kenya	1797
	birth: Jan-May; e Africa	5758
	birth: Dec-Jan; Kenya	1796
	birth: Dec-Feb; Kenya	1797
	birth: nearly yr rd; captive	5522
	birth: yr rd, peak Oct-Dec; n501, 169 f; lab	10072
Macaca arctoides	cycles: yr rd; captive	10371
	mate: yr rd; Veracruz, Mexico	3107
	birth: Mar, June-July, Nov; 63% wet season; Veracruz, Mexico	3107
	birth: Mar-June, Oct-Dec; n10; lab	6687
	birth: peak May/June; China	3386
	birth: Aug-June; n19; captive	4183
	birth: yr rd, peak July-Aug; captive	3106
	birth: yr rd, peak Aug 13/62; captive	10065
	birth: yr rd; captive	4464
	birth: yr rd; captive	7909
	birth: yr rd; captive	2783, 2785
Macaca assamensis	birth: peak Apr-June; few data; Thailand	3385
Macaca fascicularis	sperm: peak Apr-May, low Feb, July; lab	6729
	testes: peak Apr-Sept; lab	6729
	sex skin: peak Jan-Mar, low Apr-Aug, none Sept-Dec; Sumatra, Indonesia	11184
	conceive: Aug-Feb, peak Nov; n43; zoo	12120
	conceive: peak Sept-Jan, low June-Aug; lab	11064
	preg f: yr rd, peak May-June; Malaysia	5588
	preg f: Aug; n1 n Borneo, Indonesia	2313
	birth: peak Mar-May; Thailand	3385
	birth: peak Mar-June, low Sept; n111; semi wild, Japan	7954
	birth: Mar-Nov; n42; captive	4183
	birth: peak Aug-Oct; captive	3634
	birth: peak Aug-Dec; lab	10844
	birth: yr rd, peak Mar-Oct; captive	4839
	birth: yr rd, peak May-June; captive	10264
	birth: yr rd; lab	6687
	birth: yr rd; zoo	12117
	birth: yr rd; captive	2783

Macaca fuscata	spermatogenesis: Nov-Mar; Oita, Japan	10816
	spermatogenesis: June-Dec; Hiroshima, Japan	10816
	spermatogenesis: peak winter, low summer; lab	10815
	spermatogenesis: yr rd; Japan	5156
	cycles: Oct-Apr; semi-wild, Japan	7954
	cycles: Oct-May; captive, Japan	7915
	cycles: Dec-Apr; lab	429
	mate: end Jan-early Feb; Japan	7917
	mate: peak Sept-Jan; Japan	5589
	mate: Sept-Jan; captive, OR	147
	mate: Oct-Mar; Japan	10633
	mate: Oct-Apr, peak Jan-Feb; Japan	7405
	mate: Nov-Dec, none Feb-Sept; semi wild, Japan	5974
	mate: peak Dec-Jan, low Oct-Apr; Japan	5139
	mate: Dec-Feb; Oita, Japan	10816
	mate: Sept-Nov; Hiroshima, Japan	10816
	mate: fall-winter; captive	3981
	conceive: yr rd; Japan	5156
	birth: Feb-Aug, peak Mar-May; semi-wild, Japan	5974, 7954, 10667
	birth: Mar-Sept, peak May; n186; TX	3979
	birth: Mar-July; Japan	5590
	birth: peak Mar-July, peak Apr-June; Japan	7916
	birth: Apr-early June; captive	147
	birth: Apr-July, peak May; Japan	10633
	birth: May-June; Japan	5589
	birth: May-Sept; Japan	6168
	birth: June-Aug; Japan	7917
	birth: peak June-Aug, low May-Nov; Japan	5139
	repro: mid Oct-early Mar; Japan	11873
Macaca mulatta	sex maturity: peak Oct-Feb, low July-Aug; lab	401
	spermatogenesis: peak Oct, low Apr; Cayo Santiago, Puerto Rico	12060
	testes: large Aug-Dec; Puerto Rico	9417
	testes: large Sept-Nov, small Mar-May; Puerto Rico	2008
	testes: large Sept-Dec, small Jan-Aug; Cayo Santiago, Puerto Rico	9417
	testes: large Oct, small June-July; lab	11661
	testosterone: peak Mar; indoors; Sept; outdoors	4673
	ejaculation: yr rd, peak Dec, low Mar; lab	7265
	ejaculation: peak Dec-Jan, low June-July; lab	7270
	follicles: large Oct-Mar; lab	4448
	mate: Feb-May; India	8722
	mate: Apr-Sept; free ranging, Brazil	1970
	mate: none June-Sept; captive	3949
	mate: July-Jan, peak Sept-Oct; Cayo Santiago, Puerto Rico	2007
	mate: mid July-Aug, none Feb-June; Puerto Rico	5919
	mate: mid July-Dec; Cayo Santiago, Puerto Rico	6616
	mate: peak Aug-Sept; Nepal	7382
	mate: Sept-Dec; captive	3948
	mate: peak Sept-Jan; captive	11770
	mate: Sept-Mar; Puerto Rico	11219
	mate: end Sept-early Feb; India	6466
	mate: peak Oct, none June-Sept; captive	3949
	mate: Nov-Dec; India	6467
	mate: Nov-Dec; Pakistan	9175
	mate: peak Nov-Jan, none Mar; n India	10234
	ovulate: Feb-Mar, Oct-Dec; captive	8676
	ovulate: peak Sept-Nov, low May-July; captive	11772
	ovulate: Oct-Apr; captive	4451
	ovulate: peak Oct-May, low May-Sept; captive	11421
	ovulate: fall-mid winter; captive	8677
	conceive: peak Sept-Oct; Cayo Santiago, Puerto Rico	6616
	conceive: peak Sept-Dec, low Jan-Apr; lab	11064
	conceive: Oct-Nov; captive	11408

	conceive: Nov-Jan; captive	4448
	conceive: peak Nov-Feb; n15; lab	6880
	conceive: yr rd; captive	948
	conceive: yr rd; captive	8649
	birth: Jan-May, peak Mar-Apr; Puerto Rico	5919
	birth: Jan-June, peak Feb-Mar; Cayo Santiago, Puerto Rico	5920
	birth: Jan-June, peak Mar-Apr; Cayo Santiago, Puerto Rico	2007
	birth: Jan-mid July; Cayo Santiago, Puerto Rico	6616
	birth: Feb-Apr; Cayo Santiago, Puerto Rico	8975
	birth: Feb-Aug, peak Apr-May; captive	3427
	birth: Mar-May; Desechio I, Puerto Rico	7542
	birth: Mar-May, Sept-Oct; Rajasthan, India	8722
	birth: Mar-June, 71% of 31; captive, TN, NE	11216
	birth: Mar-June, 74% of 42; captive, GA	11216
	birth: Mar-Aug; captive	3950
	birth: Mar-Aug (Nov), 74% May-June; Puerto Rico	11219
	birth: Mar-Nov; n76; captive	4183
	birth: peak Mar-May; Nepal	7382
	birth: peak Mar-May, none Oct-Nov; lab	6687
	birth: mid Mar-early June; n11; captive	3948
	birth: end Mar-early Oct; Rajasthan, India	8719
	birth: end May-June; captive, Puerto Rico	11216
	birth: Apr-May, Sept-Oct; Pakistan	6766
	birth: Apr-May; India	6466
	birth: Apr-May; captive	11408, 11767
	birth: peak Apr-May, few Sept-Oct; n India	10234
	birth: Apr-Aug; Hainan I, China	5312a
	birth: May-June; free ranging, Puerto Rico	11218
	birth: peak May, July-Aug, few Nov-Mar; captive	9056
	birth: peak Oct-Jan; zoo	1230
	birth: Oct-Jan; zoos, s hemisphere	917
	birth: Oct-Mar or Oct-Dec; free ranging, Rio de Janeiro, Brazil	1970
	birth: end Dec-mid July, peak Feb-Mar; Cayo Santiago, Puerto Rico	5920
	birth: yr rd; Mar; zoo, India	4534
	birth: yr rd, peak Mar-June; n815; captive	10078
	birth: yr rd, peak Apr-May; lab	11062
	birth: yr rd, peak Apr-Sept; captive	1151
	birth: yr rd, peak late Apr-mid July; 60% May-June; captive	2721
	birth: yr rd; captive	5228
	birth: yr rd; lab	264
	birth: yr rd; zoo	12117
Macaca nemestrina	conceive: Jan-May, July-Aug, Oct-Nov; few data; Malaysia	1543
	preg f: Feb-Apr; n5; Thailand	3385
	birth: est Dec-Jan; Thailand	3385
	birth: yr rd; captive	6083
	birth: yr rd; n68; captive	4183
	lact f: Feb; n3; Thailand	3385
Macaca nigra	birth: Dec-Jan, Mar, May, June-Oct; n17; captive	4183
	birth: Dec-Oct; n41; captive	2620
Macaca radiata	mate: Sept-Oct; captive	8885
	birth: Feb-Mar; India	9982
	birth: peak Mar-May; no data Jan-Nov; captive	9973
	birth: Apr-Aug; n7; captive	4183
	birth: June-Aug; n3; zoo	44
	birth: peak spring-summer; zoo	1230
	birth: yr rd: peak Mar-July; captive	9974
	birth: yr rd; lab	11062
Macaca silenus	birth: Mar-Nov, peak Apr-May; zoo	2497
	birth: June; n1; zoo	2109
	genital swelling: peak June; India	6096
	mounting: peak June, low Mar-Apr; India	6096
	mate: yr rd; zoo	2495
	birth: Mar-Nov; zoo	2495

Macaca sinica	repro: not seasonal; Sri Lanka	8496, 8501
Macaca sylvanus	mate: mid Aug-Mar, peak Oct-Dec 80%; n1701; captive	6127, 8265
	mate: Oct-Nov, n95; Gibraltar	6711
	conceive: mid Nov-mid Dec 60%; captive	6127
	birth: Mar-Aug, Apr-May 80%; captive	8265, 8266
	birth: end Mar-July; captive	9556
	birth: peak Apr-June, none Jan-Feb; zoo	9171
	birth: Apr-July; few data; Algeria	7199
	birth: May-Sept; n145/147; Gibraltar	6711
Macaca thibetana	birth: Jan-May; Mt Emei, China	12093b
Nasalis concolor	neonates seen: June-July; Siberut I, Indonesia	10836a
Nasalis larvatus	repro: not seasonal; N Borneo, Indonesia	9667
Papio hamadryas	mate: yr rd; e Africa	6612
	mate: yr rd; few data; zoo	1230
	mate: yr rd; zoo	12120
	birth: peak Sept-Nov; captive	12118
	birth: peak Sept-Nov; Tanzania	9086
	birth: Oct-Dec; S Africa	2501
	birth: yr rd, peak mid Nov; n87; zoo	12120
	birth: yr rd, peak Dec, low Apr; n93; semi wild; Japan	7954
	birth: yr rd; forest; peak end wet season; savanna; e Africa	5758
	birth: yr rd, peak rainy season; Zaire	11253
	birth: yr rd; S Africa	10158
	birth: yr rd, except Oct-Nov; n36 over 20 mo; captive	6937
Papio leucophaeus	birth: yr rd; Cameroun	9468
	lact f: Aug; n1; s Cameroun, Benito, Equatorial Guinea	647
Papio sphinx	birth: Jan-Apr; Gabon	5457
Presbytis cristata	mate: yr rd; Malaya	7162
	preg f: Apr-May; n2; w Thailand	3385
	small yg: 1st half of the yr; Malaysia	7161
	lact f: Feb; n1; w Thailand	3385
Presbytis entellus	birth: Jan-May, Oct; Rajasthan, India	8722
	birth: Feb-Mar; nw India	4532
	birth: Mar, May-June; Himalayas, Pakistan	9175
	birth: Mar-Sept; n India	10539
	birth: Apr-May; n India	5273
	birth: yr rd, peak Dec-Apr; s India	10539
	birth: yr rd, peaks July-Aug, Dec-Mar; India	5044
	birth: yr rd; zoo	1230
Presbytis obscura	birth: Jan-Mar; Thailand	6369
Presbytis senex	mate: Oct-Jan; Sri Lanka	9361
	mate: yr rd; Sri Lanka	8619
	birth: peak Feb-Mar; sw Sri Lanka	8619
	birth: Feb-Mar; Sri Lanka	8496, 8501
	birth: May-Aug; Sri Lanka	9361
Pygathrix nemaeus	birth: peak Feb-June, low Sept-Dec; ne Vietnam	6502
Theropithecus gelada	birth: few Apr, Sept-Oct; Ethiopia	2796
	birth: Sept-Mar, July; n19; captive	4183
	birth: peak Nov-Mar, minor peak May-Aug; Ethiopia	2797

Order Primates, Family Hylobatidae, Genus *Hylobates*

Gibbons (*Hylobates* 4-13 kg) brachiate through tropical forests of southeast Asia feeding on leaves and fruit. Some species exhibit sexual dimorphism with respect to pelage color (7013a). The labyrinthine, hemochorial, discoidal placenta has an interstitial, antimesometrial implantation in the dorsal or ventral surface of the simplex placenta (2478, 4113, 7563, 10310, 10415, 11843) and may be eaten after birth (483, 1305). Neonates are furred (1305, 1810, 2109, 2489), cf 10212). Males have a baculum, seminal vesicles, penial spines (3738, 6967, 11851), and may carry yearlings (113, 1809, 2555a). Details of anatomy (2478, 5494, 6967, 11847) and birth (8629a) are available.

NEONATAL MASS		
Hylobates lar	m: 113-170 g; n1	483
	m: 380 g; r350-410; n2	9515
LITTER SIZE		
Hylobates hoolock	1 yg	988a
Hylobates lar	1 yg	988a
	1 yg ea; n2 lt, 1 f	1305
	1 yg ea; n3 lt	815
	1 yg ea; n7 lt	2109
	1 yg ea; n8 lt, 1 f	4198
	1 yg ea; n12 lt	1284
Hylobates moloch	1	557
	1 emb ea; n2 f	2313
Hylobates syndactylus	1	3665
	2 yg; n1 lt	2555a
SEXUAL MATURITY		
Hylobates concolor	f: 1st birth: 9 yr; n1	2610
Hylobates hoolock	f: 1st menses: ~7 yr	6967
Hylobates lar	f: 1st birth: 10 yr; n1	4198
CYCLE LENGTH		
Hylobates spp	follicular phase: 11-14 d	6230
	luteal phase: 10-11 d	6230
	r21-25 d; n10 cycles, 1 f	6230
Hylobates hoolock	menses: 2.6 d; sd0.66	6967
	27.83 d; sd4.07; r20-33; n6 cycles, 1 f	6967
Hylobates lar	menses: 2.38 d; sd0.54	1608
	3 wk-1 mo	483
	29.8 d; sd4.1; r21-43; n17 cycles, 2 f	1608
	30 d; r22-42; n20 f	1284
	31 d; n1 f	1305
GESTATION		
Hylobates concolor	200-214 d	8099
Hylobates lar	199 d; n1; pairing to birth	11371
	210 d; n1; menses to birth	1305
	7 mo	1284
	~7 mo; n4	9515
	215 d	2109
	7.5 mo	483
Hylobates moloch	7-9 mo	557
Hylobates syndactylus	232.0 d; r230-235; n3	4734
LACTATION		
Hylobates lar	solid food: 4 mo	2109
	dependent: 7 mo	988a
INTERBIRTH INTERVAL		
Hylobates lar	2-4 yr	4198
	~3 yr	2109
Hylobates moloch	2 yr	557
Hylobates syndactylus	289.75 d; r265-313; no lactation	4734
	619 d; yg lives	4734
	2-3 yr	1809
SEASONALITY		
Hylobates concolor	mate: end May-June; n1 pr; captive	8099
	birth: Dec; n1; captive	8099
Hylobates hoolock	birth: Nov-Feb; Naga Hills, India/Burma	7013a
	birth: Nov-Mar; s Asia	9265a
Hylobates lar	birth: early winter; s Asia	9265a
	lact f: Jan-early Apr; w Thailand	3385
Hylobates moloch	preg f: mid May-mid June; n2; N Borneo, Indonesia	2313

Order Primates, Family Pongidae

Great apes (gorillas, *Gorilla* 70-350 kg; chimpanzees, *Pan* 30-80 kg; orangutans, *Pongo* 30-140 kg) are diurnal, terrestrial to semiarboreal, often social herbivores from Africa and southeast Asia. Males are larger than females (2406, 3387, 3583, 4357, 5043, 5467, 5758, 5766, 8001, 8798, 9084, 9133, 9544). The 200-350 g, hemochorial, villous, discoid placenta, which secretes gonadotropins (10636a), implants into a simplex uterus (3737, 4000, 4113, 4842, 6183, 6633, 7554, 7563, 7714, 9293, 10310, 10415, 11843, 11847) and is usually eaten after birth (400, 697, 1068, 1889, 2407, 2962, 3584, 3888, 3890, 4385, 4583, 5786, 5787, 5788, 5793, 6583, 6758, 7938, 8798, 9084, 10399, 10400, 11087; cf 3440, 4000, 6183, 7721). Estrus does not occur (2153, 5766) after the birth of a furred, open-eyed neonate (1354, 5758, 5788, 6183, 6186, 7721, 8798, 10399; cf 6186). Males have seminal vesicles, penial spines, and a baculum (3738, 4787, 9472). The extensive endocrinological data on great apes are not reviewed here. Details of anatomy (3737, 5494, 9666a, 11847, 11851), birth (1068, 2407, 3278, 3584, 3890, 4000, 5786, 5788, 6583, 7721, 7938, 10399, 10400), mating (11495a), and milk composition (1152, 5471, 10695, 12015) are available.

NEONATAL MASS

Gorilla gorilla	1.276 kg; stillborn, premature	8914
	1.68 kg; se0.29; n4	2199
	est 1.75 kg; n1	3278
	1.8 kg; r1.7-1.9; n2	5793
	est 1.82 kg; n1	6183
	1.871 kg	3379
	f: 1.87 kg; n1	10786
	m: r1.986-2.270; n6	5791
	2.04 kg; 36 h	7721
	2.097 kg; 2 d	3536
	2.1 kg; n1	5792
	2.1 kg; n1, 21 h	6583
	f: 2.106 kg; r1.617-2.400; n9	5791
	2.14 kg; n1	9371
	2.2 kg; n1	1600
	2.2 kg; r1.6-2.6; n15	2076
	2.25 kg; n1	6583
	2.3 kg; r2.2-2.6; n3	1955
	2.375 kg; r2.20-2.55; n2	6758
	2.6 kg; n1 stillborn; ~2.6 kg; n1 live birth	4369
Pan paniscus	1.162 kg; 41 h; (infant had not eaten)	4735
Pan troglodytes	0.78 kg; r0.63-0.93; n2 stillborn, no brain, triplet birth	9293
	0.88 kg; r0.8-0.98; n3	11997
	0.89 kg; n1, 1 wk, from triplet birth	9293
	~1 kg; n5	5471
	1.2-1.5 kg; n~20	10679
	f: 1.69 kg; r1.20-2.44; n13	5614
	1.7785 kg; r1.502-2.055; n2	3379
	m: 1.81 kg; r1.35-2.20; n16	5614
	1.89 kg; r1.61-2.26; n17	12015
	m: 2 kg; n1	2962
Pongo pygmaeus	f: 0.85 kg; n1	4581
	1-2 kg; from graph	1073
	1.2 kg; small; 2.2 kg; normal	1072
	1.423 kg; r1.265-1.600; n3	9666a
	m: 1.47 kg; n1	4581
	f: 1.51 kg; r1.35, 1.67	6501
	f: 1.56 kg; sd0.39; r0.92-2.04; n14	3387
	~1.6 kg; r0.95-1.65	5878
	1.644 kg	3379
	f: 1.694 kg; r1.42-2.04; n5	9737
	1.71 kg; se0.24; n5	2199
	m: 1.74 kg; r1.59-2.015; n7	9737
	m: 1.93 kg; sd0.29; r1.59-2.26; n12	3387
	m: 2.16 kg; r2.06-2.26; n2	6501

NEONATAL SIZE		
Pan troglodytes	f: 39.1 cm; r34.3-45.7; n9	5614
	m: 40.3 cm; r37.0-44.5; n12	5614
Pongo pygmaeus	m: 32.5 cm; r32, 33; n2	11048
WEANING MASS		
Gorilla gorilla	est 4-7 kg; 6-8 mo; from graph	4816
LITTER SIZE		
Gorilla gorilla	1	9544
	1 yg	6583
	1 yg ea; n2 lt	7723
	1.04; r1-2; n56 lt	5791
	2; n1 lt	5793
	2 emb; n1 f	9283a
Pan paniscus	mode 1; r1-2	8798
	1 yg ea; n7 lt	5257
Pan troglodytes	1	10971
	1-2	11997
	mode 1; r1-2	12015
	1 yg; n1 f	1354
	1 ea; n4 lt	12128
	1.02; r1-2; n55 lt	3888, 3889
	1.04; r1-3; n291 lt	6876
	1.05; r1-2; n120 lt	8284
	1.06 yg; r1-2; n118 lt	8284
	3 yg; n1 lt	9293
Pongo pygmaeus	1	11034
	1 ea; n2 lt	3569
	1 yg; n2 lt	11048
	1 ea; n55 lt	5766
	1.1; r1-2; n28 lt	3387
	2 yg; n1 lt	320, 4581
	2 yg; n1 lt	9511a
SEXUAL MATURITY		
Gorilla gorilla	1st mate: 1st birth: 5.5 yr	5618
	f: genital swelling: 7.75 yr; r6.5-8.5; n4	4358, 4360
	f: menarche: 9 yr; n1	7945
	f: menarche: ~7 yr; r6.42-8.58; n13	3426
	f: 1st mate: median 6.33 yr; r5.75-7.1	11495b
	f: 1st mate: 6-7 yr	9544
	f: 1st mate: 7.5 yr; n5	4357
	f: 1st conceive: 7-9 yr	4360
	f: 1st preg: 9.83 yr; r6.33-15; n26	2076
	f: 1st birth: min 5.5 yr; mode 8.5	7650
	f: 1st birth: 6.75 yr	5791
	f: 1st birth: median 10.05 yr; r8.67-12.75	11495b
	f: 1st birth: 10.42 yr; r9.5-10.83	4357, 4358, 4360
	f: 2 yr	4359
	m: 1st mate: ~9 yr	9544
	m: 1st mate: 9-10 yr	11495b
	m: 1st mate: 12 yr	3426
	m: 1st mate: 15 yr	4358
	m: 1st sire: 7.67 yr	5791
	m: 1st sire: 10.08 yr; r6.33-17; n37	2076
	m: 8 yr	4357
Pan troglodytes	f: menarche: 7 yr	6171
	f: menarche: 7.7-9.3 yr	1946
	f: menarche: 8 yr	11964
	f: menarche: 8.92 yr; r7.33-10.17; n7	12047
	f: menarche: 12-15 yr	10972
	f: swelling, menarche: 6-8 yr	11177
	f: 1st cycle: 8-10 yr	12015
	f: 1st swelling: 10 yr	10972
	f: 1st mate: 11-13/14 yr	3889
	f: 1st mate: 13.25 yr; n1	10971

	f: 1st conceive: 8.2-10.7 yr; n6	1946
	f: 1st conceive: 10-10.5 yr	12047
	f: 1st preg: 13-15 yr	10970
	f: 1st birth: ~13-15 yr	10972
	f: 1st repro: >9 yr	11176
	f: 18 mo; precocious	11825
	m: sperm, testosterone: 11 yr	6879
	m: sperm: 8 yr	12015
	m: testes large, testosterone: 7-9 yr	5992
	m: testes large: 110 mo; r98-123; n3	8800
	m: 10-11 yr	2701
	m: 1st sire: 13-15 yr	3889
	m: 6-11 yr	3904
Pongo pygmaeus	f: menarche: 7-8 yr	9133
	f: menarche: 9-14 yr	2406
	f: 1st mate: 7-8 yr	4994
	f: 1st mate: 8 yr	11087
	f: 1st conceive: 8-9 yr	9133
	f: 1st birth: 8 yr	9736
	f: 1st birth: 10 yr; n1	4581
	f: 1st birth: 15 yr; n1	3583
	m: sperm: 6.4, 7.7 yr; n2	2634
	m: 1st mate: 6.6 yr	2634
	m: 1st mate: 10 yr	11087
	m: 1st sire: 7 yr	9736
CYCLE LENGTH		
Gorilla gorilla	follicular phase: 9-31 d; n4 cycles, 3 f	7731
	follicular phase: 17.1 d; ±2.0; r8-25; n5 f	2202
	follicular phase: 19.5 d; se1.0; r11-30; n21	7386
	estrus: median 1 d; r1-3; n3, 6 f	4358
	estrus: median 1 d; r1-4; n77	11495b
	estrus?: median 11 d; r1-5; n14, 5 f adolescents	4358
	luteal phase: 11-13 d; n4 cycles, 3 f	7731
	luteal phase: median 12 d; r9.5-14; n5 cycles, 5 f	7730
	luteal phase: 12.3 d; se0.3; r10-14; n21	7386
	luteal phase: 13 d; ±1.1; r8-17; n5 f	2202
	r23-48 d; n4 cycles, 3 f	7731
	r24-38 d; n1 f	7723
	median 26 d; r25-40; n7 nulliparous f, 24 cycles	3426
	r27-39 d; median 28; n5 cycles, 1 f; 38 d; 1 cycle; n1 f	4358
	28.8 d; median 28; r20-39 d; n25 cycles	11495b
	30 d; ±1.5; r25-36; n5 f	2202, 7385
	31.1 d; sd7.5; median 30.8; n56 cycles, 7 f; via tumescence	7722, 7724
	32 d; se1; r25-42; n22 cycles, 6 f	2203, 7386
	32 d; r24-44; n25 cycles, 7 f; via tumescence	7727
	32 d; r26-47; n53 cycles, 1 f	6758
	32.5 d; sd4.6; median 31.8; n25 cycles, 7 f; via urogenital cleft length	7722, 7724
	median 33 d; r21-49; n19 cycles, 5 f	7730
	33.24 d; sd4.74; se1.15; n17 cycles, 2 f	10832
	45 d; n3 cycles, 1 f	12123
	49 d; without 60, 72 d cycles; r36-72; n8 cycles, 1 f	7945
	1 mo	6183
Pan paniscus	menses 1 d	9513
	swelling phase: 22.4 d	10798
	35 d; r32-36; n3 cycles, 1 f	1068
	36.0 d; n12 cycles, 2 f	9513
	46.0 d	10798
	49 d; sd2.9; n23 intermenstrual intervals, 4 f	2213a, 2213b
Pan troglodytes	menses: 2-3 d	3889
	menses: 2-4 d	2960
	menses: 2-4 d	6171
	menses: 2-6 d	12047
	menses: 3 d	2961, 12016
	menses: 3 d	10864

	menses: 5-7 d	10850
	menses: 6 d	11177
	follicular phase: 10-32.5 d	12017
	follicular phase: 17-45 d	9072
	follicular phase: 20.4 d; se1.0; r15-25	7734
	estrus: 6 d	12016
	genital swelling: 6.5 d	11177
	luteal phase: 12-14 d	5033
	luteal phase: 12-16 d	9072
	luteal phase: 15.0 d; se0.5; r13-18	7734
	r22-43 d; n4 cycles, 4 f	5033
	r27-30 d; n6 cycles, 1 f	11964
	28 d; n2 cycles, 1 f	3066
	28 d; n2 cycles, 1 f	3997
	31 d; r27-37; n9 cycles, 3 f	7008
	32-36 d	10850
	34 d; mode 35; r32-36; n10 cycles, 1 f	4583
	median 34 d; r32-56; n6 cycles, 4 f	9072
	35 d	8443
	median 35 d; r28-53; n263 cycles, 13 f	2960
	35.4 d; se1.2; r29-43; n11 cycles, 7 f	7734
	35.5 d; r26-49; n21 cycles, 3 f	1896
	35.5 d; n50 cycles, 7 f	10864
	36 d; r31-41; n2 cycles, 1 f	2962
	36 d; sd4.82; median 35; mode 35; r29-53; n164 cycles, 11 f	2961, 12015, 12016, 12017
	median 36 d; r25-84; n46 cycles, 9 f	10970, 10972
	36.2 d; n2 f	9513
	37 d	3889
	37.28 d; se0.14; median 35; r22-187; n653 cycles, 22 f	12047
	37.6 d; n31 intermenstrual intervals	2213b
	39.7 d; r28-50; n7 cycles, 1 f	6171
	41 d	11177
Pongo pygmaeus	menses: median 2.8 d; r1-4	7726
	follicular phase 12.7 d; ±2.1	7729
	estrus: 7-10 d; n1	1679
	luteal phase: 13.5 d; ±1.3	7729
	~25 d	1679
	26.3 d; r24-28; sd1.7; n5 cycles, 5 f	7729
	26.8, 28.6 d; r24-32; n6 cycles, 2 f	1982, 3992
	28-29 d	4581
	30.1 d; median 31; r26-37; n11 cycles, 3 f	5766
	median 30.5 d; r26-32; n8 cycles, 3 f	7728
	31.3 d; r22-44; n3 cycles, 1 f; via consortship	3583
	32 d; n4 cycles, 1 f	457
GESTATION		
Gorilla gorilla	236-238 d; n1	4369
	245-247, 249-251 d; n2	7723
	251 d; n1	4979
	252 d; n1	6186
	252 or 267-268 d; n1	1955
	254 or 274 d	10832
	254.7 d; r238-274; n7	2076
	255 d	2199
	255 d; n1	10933
	256 d; n1	6887
	256-259 d; n1	9714
	257-259 d; n1	10786
	257 d; n1	697
	259.7 d; r245-280; n21; 2 outliers dropped	2153
	261 d; n1; hormonal data	6893
	8.5 mo	3426, 4357, 4358
	265-267 d; n1	6758

Pan troglodytes	~5 mo; n1	1889
	167 d; r153-179	4385
	180 d; r167-193; n31; last menses to birth	4385
	214 d; n1	4583
	226.8 d; sd13.3; r196-260; n118	8284
	227.5 d; sd12.6; median 228; r202-260; n48; detumescence to birth	7938
	228.7 d; sd12.7; median 228.5; r202-261; n48; last mating to birth	7938
	229 d; r203-244; n9	6878
	231 d; r202-261; n44	12015
	232 d; r202-261; n13	2962
	236 d; median 236; r232-261	12017
	8 lunar mo	3889
	246 d; n1	10850
	250 d	11964
	253 d; r232-278; n11; menses to birth	10864
	~9 mo	1355
Pongo pygmaeus	233 d	4106
	250 d; n1	3584
	240-259 d; n1	5766
	255-270 d; n1	11048
	255-275 d	8798
	261 d; n1	400
	263 d; n1	4000
	270.7 d; mode 270; n15	2153
	275 d	457
	294.5 d; r292-297; n2	6500
LACTATION		
Gorilla gorilla	solid food: 2.5 mo	9544
	solid food: 3 mo; regularly 5-6 mo	8798
	solid food: 12-14 wk; n1	6186
	wean: est 6-7 mo	4816
	wean: 8 mo	9544
	wean: 2 yr	8798
Pan paniscus	solid food: 3 mo; regularly 5-6 mo	8798
Pan troglodytes	solid food: 4-6 mo	3888
	solid food: 6 mo	9084
	solid food: 10 mo	10679
	wean: 15 d-10 mo	3904
	wean: 10-18 mo	6342
	wean: 18 mo	10679
	wean: 1-2 yr	9084
	wean: 1.5-2.5 yr	3888
	wean: 2-4 yr	11176
	wean: 64.6 mo; r50-86; n5 m	8800
	wean: 63.2 mo; r50-72; n6 f	8800
Pongo pygmaeus	solid food: 6-9 mo; regularly 1 yr	8798
	wean: 3 yr; n1	1679
	wean: 3-4 yr	8798
	wean: 1-2.5 yr	5766
INTERBIRTH INTERVAL		
Gorilla gorilla	3 yr	9544
	3-4.92 yr; shorter if yg die	4360
	3.5 yr; n9	4357
	median 3.83 yr; r3-4; n6 f; yg live	4358
	3.92 yg; r3.0-7.25; n26 ibi; yg live	9969a
	4.16 yr; median 4; r2.33-6.42; n16 ibi, 13 f; yg live	9969a
	median 4.42 yr; r3-7; n8 f; yg die	4358
Pan paniscus	4.5 yg; yg survives	5257
Pan troglodytes	14.5 mo	3904
	2.5-3 yr	3888
	min 3 yr	11176
	3-4 yr	9084

	4.38 yg; r3-7; n13 intervals, 6 f	10537
	4.5-7.5 yr	3889
	5-6 yr; n3 f; yg lives	7924
	5.67 yr; r4.58-7.58; n15 intervals, 12 f	10971
	5.83 yr; r4.33-7.5; n8 f	10970, 10972
	6.0 yr; r5.16-7.08; n7	11415
Pongo pygmaeus	2.5-3 yr	4994
	>5 (6-7)? yr	3583
SEASONALITY		
Gorilla gorilla	birth: yr rd; Virunga Volcanoes, Albert Nat Park, Zaire	9544
	birth: yr rd; zoo	2076
	repro: not seasonal; central Africa	8798
	repro: not seasonal; e Africa	5758
Pan paniscus	birth: not seasonal; Congo R, Zaire	8798
Pan troglodytes	birth: yr rd; captive	7938
	birth: yr rd; lab	6876
	repro: not seasonal; w, equatorial Africa	8798
Pongo pygmaeus	birth: Mar, May; n2; captive	1679
	birth: May; n1; captive	400
	repro: not seasonal; Indonesia	8798

Order Primates, Family Hominidae, Genus *Homo*

Humans, *Homo sapiens*, are bipedal, terrestrial omnivores with a global distribution. Males are larger than females. Females have 2 mammary glands and a simplex uterus. Menstruation and concealed ovulation are characteristic. Neonates are born with open eyes. Review of the extensive literature (5891, 12009a) on human reproduction is outside the scope of this compendium.

NEONATAL MASS		
Homo sapiens	<2500 g; low	2040
	2909-3249 g	8337
	3179 g; se105; n12	1078
	f: 3200 g; sd550; n6693	5541
	3250-3600 g	9285
	m: 3300 g; sd600; n7037	5541
	4070 g; se109; n4 large	1078
LITTER SIZE		
Homo sapiens	mode 1; 2 rare	8244
SEXUAL MATURITY		
Homo sapiens	f: menarche: 12-14 yr	11149
	f: menarche: 13.533 yr; r11.044-16.310; n250	3040
CYCLE LENGTH		
Homo sapiens	28.7 d; sd3.74; 26-28 45%; r of x 17-58; n770 f	4154
	30.4 d; 18-42 97%; r11-144; n747 cycles	3343a
	33.9 d; sd14.28; mode 28; 17-45 87%; r7-256; n3140 cycles, 250 f	3040
GESTATION		
Homo sapiens	259 d	2040
	279.66 d; sd15.41; n7037 f offspring	5541
	280.60 d; sd14.37; n6693 m offspring	5541
	43 wk	1078
LACTATION		
Homo sapiens	0.5-3 yr	8677a
INTERBIRTH INTERVAL		
Homo sapiens	15 mo; r8-38; n50 f no lactation	9519
	27 mo; r8-44; n209 f with lactation	9519
SEASONALITY		
Homo sapiens	birth: Mar-June >50%; Kenya	6400
	birth: yr rd; worldwide	2093